

THE JOURNAL OF GEOLOGY

A Semi-Quarterly Magazine of Geology and
Related Sciences.

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VOLUME I

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CHICAGO

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THE JOURNAL OF GEOLOGY

JANUARY-FEBRUARY, 1893.

ON THE PRE-CAMBRIAN ROCKS OF THE BRITISH ISLES.

During the last twenty years much has been written about the "pre-Cambrian" rocks of the British Isles. Unfortunately when attention began to be sedulously given to the study of these ancient formations, the problems of metamorphism were still a hundred fold more obscure than they have since become; the aid of the microscope had not been seriously and systematically adopted for the investigation of the crystalline schists, and geologists generally were still under the belief that the broad structure of these schists could be treated like those of the sedimentary rocks, and be determined by rapid traverses of the ground. We have now painfully discovered that these older methods of observation were extremely crude, and that the work performed in accordance with them is now of little interest or value save as a historical warning to future generations of geologists. Geological literature has meanwhile been burdened with numerous contributions which remain as a permanent incubus on our library shelves.

It may serve a useful purpose at the present time in possibly aiding those who are engaged in the study of the oldest rocks of North America, if I place before them, as briefly as possible, the main facts which in my opinion have now been satisfactorily proved regarding the corresponding rocks of Britain, and if I indicate at the same time some of the more probable inferences in those cases where the facts, at present known, do not warrant a definite conclusion.

It is obvious that in any effort to establish that a group of rocks is older than the very base of the sedimentary fossiliferous formations, we must somewhere find that group emerging from under the bottom of these formations. Until lithological characters are ascertained to be so distinctive and constant as to be comparable to fossil evidence for purposes of stratigraphical identification, we should not assume that detached areas of older rocks rising amid Palæozoic, Secondary or Tertiary formations are

pre-Cambrian. We should, if possible, begin at the bottom of the Palæozoic systems and work backward, tracing each successive system or group as these rise from under each other, until we arrive at what appears to be the oldest traceable within the region of observation. It is clear that in the present state of knowledge we have no satisfactory means of identifying such successive systems in widely separated countries. All that can be attempted in the meantime is to ascertain the special types in each region, and to point out their general resemblances or contrasts to those of other regions. It is better to avoid confusion by refraining from applying the stratigraphical names adopted for the oldest rocks of one region to those of another geographically remote, though we may hope that eventually it may be possible to work out the equivalence of these local names.

In the British Isles, by much the most important region for the study of the oldest rocks is to be found in the north-west Highlands of Scotland. The very basement strata of the Cambrian system are there traceable for a distance of more than 100 miles, reposing with a strong unconformability upon all rocks of older date. They consist of dolomitic shales with *Olenellus*, resting upon a thick group of quartzites, full of annelid tubes. One of the most remarkable features of these ancient strata is the persistence of their component bands or zones which, though sometimes only a few feet thick, can be traced throughout the whole tract of country just referred to. For the study of the pre-Cambrian rocks this is an important point, for we can be quite certain that even where fossil evidence locally fails, the same basement members of the Cambrian system are persistent and lie directly upon the pre-Cambrian series.

Lewisian Gneiss. Ever since the researches of Murchison and Nicol in the north-west of Scotland, it has been known that two distinct systems of rock underlie the quartzites to which I have just alluded. Murchison regarded the upper of these as of Cambrian age, while he assigned the unconformable quartzites and limestones above it to the Lower Silurian period. But the recent discovery of the *Olenellus* zone intercalated conformably between the quartzites and the overlying limestones may be regarded as proving that all the rocks which underlie the quartzites and are separated from them by a strong unconformability must be pre-Cambrian. It is thus established beyond any reasonable doubt that two great pre-Cambrian systems of rock exist in the north-west of Scotland.

These two systems differ so entirely from each other that their respective areas can be defined with minute accuracy. The uppermost consists chiefly of dull reddish sandstones with conglomerates, and especially towards their base in Rosshire, some bands of dark grey shale, the whole having a thickness of at least 8,000 or 10,000 feet, though as both the base and the top of the series are marked by strong unconformabilities, the whole original thickness of deposits is nowhere seen. As these rocks are well developed around Loch Torridon, they were named by Nicol the Torridon Sandstone—a designation which has more recently been shortened into "Torridonian." The lower system is mainly composed of various foliated rocks which may be embraced under the general term "gneiss." These masses present the usual characters of the so-called "fundamental complex", "Urgebirge," or "Archæan Series" of other countries. The contrast between the thoroughly crystalline, gnarled, ancient-looking gneisses below, and the overlying, nearly horizontal Torridonian conglomerates, sandstones, and shales, which are largely made out of their debris, is so striking that every observer feels persuaded that in any logical system of classification they can not be both placed in the same division of the geological record. They are certainly both pre-Cambrian, but they must belong to widely separated eras, and must have been produced by entirely different processes. If it is proposed to regard the gneisses as "Archæan," we must refuse to include the Torridonian strata in the same section of pre-Cambrian time. But so much uncertainty exists as to the application of this term Archæan, examples are so multiplying wherein what was supposed to be the oldest and truly Archæan rock is found to be intrusive in rocks that were taken to be of much younger date, and there are such slender grounds for correlating the so-called Archæan rocks of one country with those of another, that I prefer for the present, at least, not to use the term at all. Let me very briefly state some of the main characteristics of the two sharply contrasted rock-systems of the north-west of Scotland.

The oldest gneiss of that region was originally called "Lewisian" by Murchison, from its large development in the Island of Lewis, and I think it would be, for the present at least, an advantage to retain this geographical appellation. At first this "fundamental gneiss" was thought to be a comparatively simple formation, and the general impression probably was that it should be regarded as a metamorphic mass, produced mainly from

the alterations of very ancient stratified rocks. Its foliation-planes were believed to be those of original deposit which by terrestrial disturbance had been thrown into numerous plications and corrugated puckerings. But a detailed study of this primeval rock has revealed in it a far more complicated structure. The supposed bedding-planes have been ascertained to have nothing to do with sedimentary stratification, and the gneiss has been resolved into a complex series of eruptive rocks, varying from a highly basic to an acid type, and manifestly belonging to different times of extrusion. With the exception of one district, to which I shall immediately refer, no part of the whole region yet examined has revealed to the rigid scrutiny of my colleagues of the Geological Survey, any trace of rocks which can be regarded as probably of other than igneous origin. It is true that our researches have been hitherto confined to the mainland of Scotland, the large area of the Outer Hebrides, which consists of similar gneisses, remaining to be explored. It is therefore possible that indisputable evidence of an ancient sedimentary series through which the gneiss was originally protruded, may yet be discovered in the unexplored islands. But taking the gneiss as at present known in Sutherland and Rosshire, we find it to be generally coarse in texture, rudely foliated, and passing sometimes into massive types in which foliation is either faintly developed or entirely absent. Much of this gneiss is considerably more basic than the more typical rocks to which the term gneiss was formerly restricted. It consists of plagioclase felspar with pyroxene, hornblende, and magnetite, sometimes with blue opalescent quartz, and sometimes with black mica. These predominant minerals are segregated in different proportions in the different bands, some bands consisting mainly of pyroxene or hornblende, with little or no plagioclase, others chiefly of plagioclase, with small quantities of the ferro-magnesian minerals and quartz, others of plagioclase and quartz, others of magnetite. This separation of mineral constituents can hardly be attributed to mere mechanical deformation. It rather resembles the segregation layers which may be studied in intrusive sills and other deep-seated masses of eruptive material, and which are obviously due to a process of separation that went on while the igneous magma was still in a liquid or viscous condition. At the same time it is manifest that extensive dynamical changes have affected the rocks since the appearance of this original banded structure.

There is further evidence that beside the original eruptive masses, which for want of any means of discriminating their relative dates of protrusion must in the meantime be regarded as belonging to one eruptive period, other portions of igneous material have been subsequently and at successive epochs, after the first mechanical deformations, injected into the body of the original gneiss. These consist of dykes of basalt and dolerite, followed by still more basic peridotites and picrites, and lastly by emanations from a distinctly acid magma in the form of granites. The oldest or doleritic dykes form a wonderful feature in the gneiss, from their abundance, persistence and uniformity of trend in a west-northwest direction. They have no parallel in British Geology until we reach the crowded dykes of older Tertiary time.

Throughout this remarkable complex of eruptive material, though its different portions present many features that may be compared with those of intrusive bosses and sheets belonging to later geological periods, there is no trace of any superficial volcanic manifestation. No tuffs or agglomerates or slaggy lavas have been detected, such as might serve to indicate the ejection of volcanic materials to the surface. All the phenomena of the Lewisian gneiss point to the consolidation of successively protruded portions of eruptive material at some depth within the crust.

Nevertheless it may yet be possible to show that these deep seated masses have been injected into rocks of older date and of sedimentary origin, and that they have communicated with the surface in true volcanic eruptions. I have already alluded to one limited area where various rocks exist, distinctly different from the prevalent types in the Lewisian gneiss. In the area which is traversed by the long valley of Loch Maree in western Ross-shire, there occur clay-slates, fine mica schists, graphitic schists, and saccharoid limestones. These rocks remind us of some of the prevalent members of a series of metamorphosed sediments. The minerals enclosed in the marbles are just such as might be expected in the metamorphic aureole of a granite boss, piercing limestone. But the relations of this group of rocks to the ordinary gneiss of the region are not quite so clear as could be desired, though they seem to point to these rocks being surrounded by and enclosed within the gneiss.

The detailed field-work of the officers of the Geological Survey has made known the remarkable amount of mechanical deformation which the

various rock-masses composing the Lewisian gneiss have undergone. These rocks have been compressed, crushed, and drawn out, until what were originally massive crystalline protrusions have been converted into perfect schists. The dykes of dolerite have been transformed into hornblende-schists and the granitic pegmatites have been reduced to a kind of powder which has been rolled out so as to simulate the flow-structure of a lava. There is evidence that most, if not all, of this dynamical change was effected long before the deposition of the Torridonian series, for the latter rests in nearly horizontal sheets, with a strong unconformability upon the crushed and sheared gneiss.

Torridon Sandstone. This group of rocks covers only a limited area in the north-west of Scotland, but it must once have spread over a far more extensive region. It reaches a thickness, as I have said, of 8,000 or 10,000 feet, and consists almost wholly of dull, purplish-red sandstones, often pebbly, and bands of conglomerate. Dark grey shales, already alluded to as occurring towards the base of the series, are repeated also in the highest visible portion, and have yielded tracks of what seem to have been annelids and casts of nail-like bodies which may have been organic. I have said that the Torridonian deposits which were classed by Murchison as Cambrian, have been proved by the discovery of the *Olenellus* zone in an unconformable position above them, to be of pre-Cambrian age. Except along the line of disturbance to which I shall immediately refer, these strata are quite unaltered. Indeed, in general aspect they look as young as the old red sandstones with which Hugh Miller identified them. It is at first hard to believe that such flat undisturbed sandstones are of higher antiquity than the very oldest Palæozoic strata which are so generally plicated and cleaved.

The interval of time between the deposition of the Torridon Sandstone and of the overlying Cambrian formations must have been of enormous duration, for the unconformability is so violent that the lowest Cambrian strata, not only transgressively overspread all the Torridonian horizons, but even lie here and there directly on the old gneiss, the whole of the intervening thick mass of sandstone having been there removed by previous denudation. At Durness, in the north of Sutherland, about 2000 feet of Cambrian (possibly in part Lower Silurian) strata can be traced, the lower portion consisting of quartzites, the central and upper parts of various limestones, sometimes abundantly fossiliferous. Nowhere else in the north

of Scotland can so thick a mass of early Palæozoic rocks be seen. Elsewhere the limestones have been in large measure replaced by a complex group of schistose rocks which rest upon the Cambrian strata, and like them dip, generally at gentle angles, towards the east. It was the opinion of Murchison, and was commonly admitted by geologists, that these overlying schists represented a thick group of sediments, which, originally deposited continuously after the limestones, had been subsequently altered into their present condition by regional metamorphism. They were variously named the "Eastern schists," the "younger gneiss," the "gneissose and quartzose flagstones." Nicol, who at first shared the general opinion regarding them, afterwards maintained that they did not belong to a later formation than the limestones, but were really only the old gneiss, brought up again from beneath by enormous dislocations and over-thrusts. We now know from the labors of Professor Lapworth and the officers of the Geological Survey, that Murchison and Nicol had each seized on an essential part of the problem, but that both of them had missed the true solution. Murchison was in error in regarding his younger gneiss as a continuous sequence of altered sedimentary rocks conformably resting on the Cambrian (or to use his terminology, Lower-Silurian) formations. But he sagaciously observed the coincidence of dip and strike between the schists and sedimentary rocks below them and inferred that this coincidence, traceable for many leagues, proved that the metamorphism which had given these schists their structure must have taken place after the deposition of the Durness limestones. Nicol, on the other hand, with great insight recognized that there was no continuous sequence above those limestones, but that masses of the old gneiss had been thrust over them by gigantic faults. But he failed to see that no mere faults would account for the coincidence between the structural lines just referred to in the Cambrian strata, and in the overlying schists, and that the general tectonic structures and lithological characters of the eastern schists differed in many respects from those of the Lewisian gneiss.

The problems in tectonic geology presented by the complicated structures of the northwest of Scotland have been ably worked out by the officers of the Geological Survey, to whose report in the *Quarterly Journal of the Geological Society* for 1888, I would refer for full details. It has been shown that, besides stupendous dislocations and horizontal displacements,

the rocks have been cut into innumerable slices which have been driven over each other from the eastward, while at the same time there has been such a general shearing of the whole region that for many hundreds of square miles the original rock-structures have been entirely effaced, and have been replaced by new divisional planes, which, when they approach the underlying Cambrian strata, are roughly parallel with the bedding planes of these strata.

In this region, therefore, we have striking proofs of a stupendous post-Cambrian regional metamorphism. But there is still much uncertainty regarding the geological age of the rocks which have been affected by it. There can be no doubt that large masses of the old gneiss, torn up from below, have been thrust bodily westward for many miles, and are now seen with their dykes and pegmatites resting on the Durness limestones and quartzites. It is equally certain that in other districts huge slices of the Torridon sandstones have been similarly treated. But where all trace of original structure has disappeared, we have, as yet, no means of definitely determining from what formation the present eastern schists have been produced. The ordinary gneissose and quartzose flagstones do not appear to me to be such rocks as could ever be manufactured by any chemical or mechanical process out of the average type of Lewisian gneiss. I have long held the belief that they were originally sediments, but whether they represent altered Torridon Sandstone, or some clastic formations which may have followed the Durness limestones, but which have been everywhere and entirely metamorphosed, remains for future discovery. For my present purpose, it is sufficient to observe that, in the meantime, as we can not be sure of the origin of most of the rocks, which, between the West Coast and the line of the Great Glen, have been subjected to a gigantic post-Cambrian regional metamorphism, it seems safest to exclude them from an enumeration of the pre-Cambrian rocks of Britain.

Dalradian. East of the line of Great Glen, which cuts the Scottish Highlands in two, another group of crystalline schistose rocks is largely developed. It consists mainly of what were undoubtedly originally sedimentary deposits, though they are now found in the form of quartzites, phyllites, graphitic schists, mica-schists, marbles, and various other foliated masses. With them are associated numerous eruptive rocks, both acid and basic, sometimes still massive and easily recognizable as intrusive,

sometimes more or less distinctly foliated and passing into different gneisses, hornblende-schists, chloritic-schists, etc. Though it is not always possible in such a series of metamorphic rocks to be certain of any real chronological order of succession, those of the Highland tracts have now been mapped in detail over so wide an area, that we are probably justified in believing that a definite sequence can be established among them. These masses must be many thousand feet thick. Their succession and association of materials are so unlike those of any of the known older Palæozoic rocks of Britain, that they can hardly be the metamorphosed equivalents of any strata which can be recognized in an unaltered condition in these islands. Some traces of annelid casts have been found in the quartzites, but otherwise the whole series has remained entirely barren of organic remains.

What then is the age of this important series? I must confess that in the meantime I can give no satisfactory answer to this question. I have proposed, for the sake of distinction and convenient reference, to call these rocks "Dalradian." Murchison supposed them to be a continuation of his Durness quartzites, limestones, and "younger gneiss." His belief may still prove to be in some measure well founded. But at present we have no means of deciding whether the quartzites and limestones of the Central Highlands are the more altered equivalents of the undoubtedly Cambrian strata of the north-west. It is possible that in the vast mass of metamorphosed rocks constituting the wide stretch of country from the northern headlands of Aberdeen to the south-western promontories of Argyllshire, there may be portions of the old Lewisian gneiss, tracts of highly altered Torridon sandstone, belts of true counterparts of the Cambrian quartzites and limestones of Durness, and, what should not be forgotten, considerable portions of some later sedimentary series which may have followed these limestones, but which, by the great dislocations already referred to, have disappeared from the north-west of Scotland. We are gradually learning more of these rocks, as the detailed mapping of them by the Geological Survey advances, and when the ground on either side of the Great Glen is surveyed, it may be possible to speak with more certainty regarding their true geological relations.

A glance at a geological map of the British Isles will show that the metamorphic rocks of the south-western Highlands of Scotland are prolonged into the north of Ireland, where they spread over a region many

hundred square miles in extent. They retain there the same general character and present the same difficult problems as to their true stratigraphical relations. Quite recently, however, a new light seems to have arisen upon these Irish rocks. My colleagues on the Irish Branch of the Geological Survey have detected several detached areas of coarse gneisses, which in many respects resemble parts of the Lewisian gneiss of north-west Scotland. In some cases these areas lie amidst or close to "Dalradian" rocks, but with that obstinacy, which so tries the patience of the field-geologist, they have persistently refused to disclose their true original position with regard to these. Some fault, thrust-plane, tract of boulder-clay or stretch of bog is sure to intervene along the very junction-line where the desired sections might have been looked for. There can be little doubt that a strong unconformability exists between them. A close examination of the ridge of old gneiss in Tyrone and Fermanagh showed me that though the actual basement-beds of this Dalradian series could not be seen resting on the coarse gneiss, the lithological character, and tectonic arrangement of this series are only explicable on the supposition of a complete discordance between it and the gneiss. As these two groups of rock have never been found in close proximity in Scotland, and as the determination of the true age of the Dalradian series is a question of such great stratigraphical importance in the general mapping of the United Kingdom, I requested Mr. A. McHenry, of the Geological Survey of Ireland, to continue the tracing of the mutual boundaries of the old gneiss of the Ox Mountains and the Dalradian series in County Mayo. He informs me that he has found in that series a conglomerate full of blocks of the old gneiss, and resting in one locality apparently unconformably upon it. If this observation is confirmed it will finally set at rest the relative position of the coarse massive gneiss and some portion, at least, of the Dalradian series. Of course there is no absolute proof that the coarse gneisses of Ireland are really the equivalents of the Lewisian masses which they so closely resemble. But there is a strong presumption in favor of their identity.

In England and Wales many detached areas of rock have been claimed as pre-Cambrian, and successive formations have been classified among them. I have already dealt in part with this question, and without attempting here to review the voluminous literature of the subject, I will content myself

with stating briefly what seems to me to have been established on good evidence.

There can not, I think, be now any doubt that small tracts of gneiss, quite comparable in lithological character to portions of the Lewisian rocks of the north-west of Scotland, rise to the surface in a few places in England and Wales. In the heart of Anglesey, for example, a tract of such rocks presents some striking external or scenic resemblance to the characteristic types of ground where the oldest gneiss forms the surface in Scotland and the west of Ireland. In the Malvern Hills another small knob of somewhat similar material is obviously far more ancient than the Cambrian rocks of that locality. There may possibly be still some further exposures of similar rocks in the south of England, as for instance in southern Cornwall. In Anglesey a series of schists, quartzites and limestones has been included by Mr. J. F. Blake with the coarse gneiss above referred to, and a thick higher group of slates in what he terms the "Monian" system. These schists, quartzites and limestones present a close resemblance to the Dalradian series of Scotland and Ireland, and the quartzites, like those of the Highlands, contain worm-burrows. The coarse gneiss, as I have said, may be compared in general character with parts of the Lewisian rocks, so that we seem to have here, as in Ireland, two groups of schistose rocks, and both of these must be much older than the unaltered Cambrian strata which lie above them.

Along the eastern borders of Wales, there is an interrupted ridge of igneous rocks which were originally supposed to have broken through the older Palæozoic formations, but which now, owing mainly to the labors of Dr. Callaway and Professor Lapworth, are shown to be older than the base of the Cambrian system. These rocks consist of spherulitic and perlitic felsites, with volcanic breccias and tuffs. They are undoubtedly older than the *Olenellus* zone. Though the evidence is not quite satisfactory, they may not impossibly lie at the base of a vast mass of sedimentary rocks forming the ridge of the Longmynd. In that case the whole of the Longmynd succession with the volcanic group at its base must be pre-Cambrian and lie unconformably below the *Olenellus* zone. Dr. Callaway has proposed the name "*Uriconian*" for this volcanic group, while the sedimentary series has been termed "*Longmyndian*." On the supposition that the unconformability is established, there would here be a vast mass of stratified and partly

erupted material forming a pre-Cambrian formation. Whether in that case any portion of this English series is the equivalent of the Torridonian rocks of Scotland remains to be determined. The northwestern part of the Longmynd ridge is made of red sandstones and conglomerates, which certainly resemble the Torridonian rocks of Ross and Sutherland.

At the base of the Cambrian rocks in Wales, Dr. Hicks has described a marked volcanic series under the name of "Pebidian," which he claims as pre-Cambrian, alleging that it is separated from the Cambrian system by an unconformability, and a band of conglomerates. I have carefully studied the evidence on this ground, and have come to the conclusion that there is no unconformability at the line in question, but that the ordinary Cambrian strata graduate downwards into the volcanic group and can not be disjoined from it. I therefore regard the so-called "Pebidian" as merely marking the duration of a volcanic period in early Cambrian time.

It will thus be seen that according to my view the unmistakably pre-Cambrian rocks of Britain consist of, first and oldest, the Lewisian gneiss; second, the Torridonian sandstones and conglomerates. The Uriconian and Longmyndian formations may prove to be in part or in whole equivalents of the Torridonian. The Dalradian rocks have not yet had their position determined. They may possibly mark a distinct pre-Cambrian series, but it seems quite as probable that they are only a metamorphic complex in which Archæan, Torridonian and Cambrian, or even Lower Silurian rocks are included.

SIR ARCHIBALD GEIKIE,
Director-General of the Geological Survey of Great Britain and
Ireland.

ARE THERE TRACES OF GLACIAL MAN IN THE TRENTON GRAVELS?

In a paper published in *Science*, Nov. 25. 1892, I undertook to study the evidence relating to paleolithic man in the eastern United States from a new point of view,—that furnished by certain recently acquired knowledge of the contents of quarries and shops where modern aboriginal flaked implements were made. It was shown that all rudely flaked forms could be sufficiently accounted for without the necessity of assuming a very rude state of culture, and that any people, paleolithic or neolithic, would in roughing out blades—the principal product of the flaking process—produce precisely these forms and in great numbers as refuse. It further appeared that the finding of these objects in sporadic cases in glacial gravels or in any formation whatsoever, could not be considered as proving or tending to establish the existence of a particular grade of stone-age culture for the region in which the formation occurs, since they may as readily pertain to a neolithic as to a paleolithic status. It was conclusively shown that no worked stone that can with reasonable safety be called an implement has been reported from the gravels, and that it is therefore clearly useless, not to say unscientific, to go on enlarging upon the evidence of an American paleolithic period and multiplying theoretic details of its culture.

I now propose to review briefly the question of the age of our so-called paleolithic implements, the questions of the *grade* of a given feature of culture and of the *age* or chronologic place of that culture being very properly treated separately, as they depend for their support upon distinct classes of evidence. During the past summer, 1892, certain important items of new evidence have been discovered bearing upon the question of the occurrence or non-occurrence of rudely flaked stones or of any artificial objects whatsoever in the normal gravels of the Delaware Valley, and it therefore becomes necessary to examine somewhat critically such of the published evidence as seems to be seriously affected by these recent observations.

It may be stated in beginning that no one disputes the glacial age of the Trenton gravels. The question to be discussed is simply this,—is the evidence satisfactory that works of art have been found in these gravels? Nothing else need be asked or answered. I do not take up this subject because I love controversy; disputation is really most distasteful to me. It happens that under the Bureau of Ethnology of the Smithsonian Institution I have been assigned to the work of making a survey of the archeology of the Atlantic coast region in which large areas, especially in states south of Mason and Dixon's line, remained almost untouched by investigators, and two years have been consumed mainly in these southern areas. But there are questions that refuse to be confined to definite geographic limits, and evidence secured in one section is sometimes found to bear so directly and forcibly upon problems pertaining primarily to other sections that the student of these problems must perforce become a free lance, and unhesitatingly enter any province promising results of value, howsoever fully occupied it may be by other investigators. One of the most interesting and important questions growing out of the study of American archeology has, as we have seen, arisen in the Delaware Valley, and the turn taken by some of my work in the south and west is such that I cannot pass this question by without consideration. The necessity of taking up the subject of glacial man became more and more apparent as the years passed on, and people continued to say to me, "You must go to Trenton; we are not satisfied with the present status of the question there; the evidence arrayed in favor of the theory of a paleolithic gravel man needs critical examination."

The difficulty of taking up and re-examining evidence, of which the record only remains, is, however, very great, since in most cases the evidence rests upon or consists of field observations, and these cannot be recalled or repeated, and there is absolutely no means of testing directly the value of what is recorded. One may seek either to verify or to discredit the promulgated theories, but years of search may fail to produce a single new item of evidence bearing decisively upon the subject. It is possible that at one period numerous finds of implements should be reported from certain portions of the gravels, and that afterwards the whole remaining body of these formations should be worked over and searched without securing a trace of art; yet this latter evidence, being negative, need not necessarily be

considered sufficient to overturn the original positive evidence if that happens to be of a high class. There is not the least doubt, however, that positive evidence may be so impaired by various defects and inconsistencies, that, unsupported by renewed and well verified observations, it will finally yield to the negative forces; and if the theories of a gravel man in the eastern United States, howsoever fortified by accumulated observations, are not really properly supported in every way, they are bound in time to fall to the ground. All I can reasonably hope to do now is to have the evidence relating to glacial man placed on trial, and so fully examined and cross-examined that those who accept gravel man need not longer do so blindly without knowing that there are two sides to the question, and those who do not accept him may know something of the reasons for the belief that is in them.

The evidence employed to prove the presence of a race of men in the Delaware Valley in glacial times is confined almost wholly to the alleged discovery of rude implements in the glacial gravels. Practically all the evidence has been collected by Dr. C. C. Abbott, and upon his skill as an observer, his faithfulness as a recorder, his correctness of judgment and his integrity of character, the whole matter stands. Many visitors, men of high repute in archeology and geology, have visited the site, but the observations made on such occasions appear not to have been of a nature to be of great value in evidence, the finds being doubtful works of art or not having properly established relationships with the gravels in place. In the discussion of gravel man in eastern America a wide range of objects and phenomena has been considered, but the real evidence, upon which the theory of an ancient race and a peculiar culture must depend, is furnished by a hundred pieces—more or less—of rudely flaked stones said to have come from the gravels in place. And now what can be said with reference to this series of flaked stones further than that they are reported by the collector to have been found in the gravels at definite stated depths? I have elsewhere shown that they are not demonstrably implements in any case, that they are identical in every respect with the quarry-shop rejects of the American Indian, that they do not closely resemble any one of the well established types of European paleolithic implements, and that they are not a sufficient index of a particular stage of culture. I shall now present such

reasons as there may be for the belief, held by many, that they were not really found in the undisturbed glacial gravels.

It is generally understood that the earliest reported gravel finds of importance were made on the banks of Assanpink creek within the city limits of Trenton, where the gravels to a thickness of twenty feet or more were exposed in a railway cutting. Later the river bluff near the lower end of the city, where the gravels were exposed to a depth of from twenty-five to forty feet, yielded large numbers. These two sites, so far as I can learn, furnished at least three-fourths of the finds in place. Other specimens were found singly in slight natural exposures, and in excavations for cellars, sewers, etc., at various points within the city limits.

The river bluff was for a considerable period the favorite hunting ground of the searchers for rudely flaked stones, and many specimens were collected. The gravels were exposed in a steep, nearly straight bank, several hundred yards in length, the base of which was washed by the river. There can be no question that Dr. Abbott and others have found shaped objects of various classes upon and in the face of this river bluff, and the visitor to-day, although the bluff is now buried almost completely under city refuse, will hardly fail to find some rudely flaked form in the deeper gullies or upon the narrow river bank or beach at the base. Dr. Abbott explicitly states^[1] that he obtained certain of these specimens from the gravel outcrops, and that they were not in talus formations, but in undisturbed deposits. How then is it possible to do otherwise than accept these statements as satisfactory and final?

^[1] Abbott, C. C. Primitive Industry, pp. 493-510.

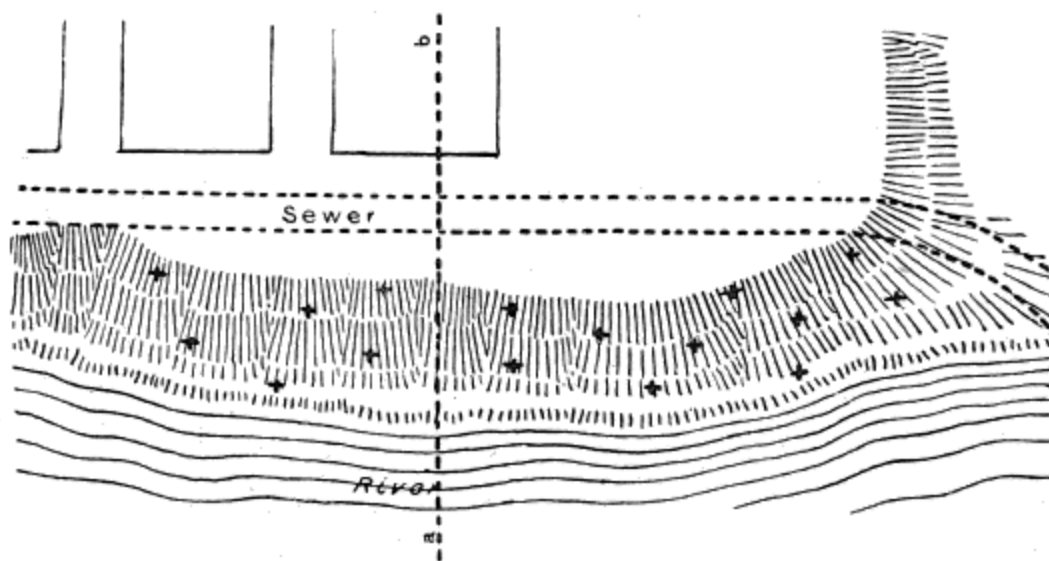


FIG. 1. Sketch map of the Trenton bluff, showing the relation of the sewer trench to the "implement" yielding slope.... a-b section line, [Fig. 2](#).

Very recently, however, fortunate circumstances have brought the evidence furnished by this site again within our reach, thus enabling us to re-open the discussion under favorable conditions. What I had for some time desired to do in this case was, what I had already done at Piny Branch, D. C., and at Little Falls, Minn., to open a trench into the face of the bluff, and thus secure evidence for or against the theory of a gravel man. This measure was, however, rendered impracticable by the occupation of the bluff margin by a city street; but it happened last summer that the city authorities, desiring to improve the sanitary condition of the city, decided to open a great sewer through this very bluff to get a lower outlet to the river. A trench twelve feet wide and some thirty feet deep, the full depth of the exposed gravels, was carried along the bluff just inside of its margin, opening out into the river at the point where the bluff turns toward the north-east. It was a trenching more complete and more satisfactory than any of which I had ever dreamed. At no point for the entire length of the bluff did the excavation depart more than forty feet from the line of the terrace face—from the upper margin of the slope upon which such plentiful evidence of a supposed gravel man had been obtained. The accompanying map and section, Figs. 1 and 2, will indicate the location of the trench, and show the exact relations of the natural and artificial exposures of the gravels.

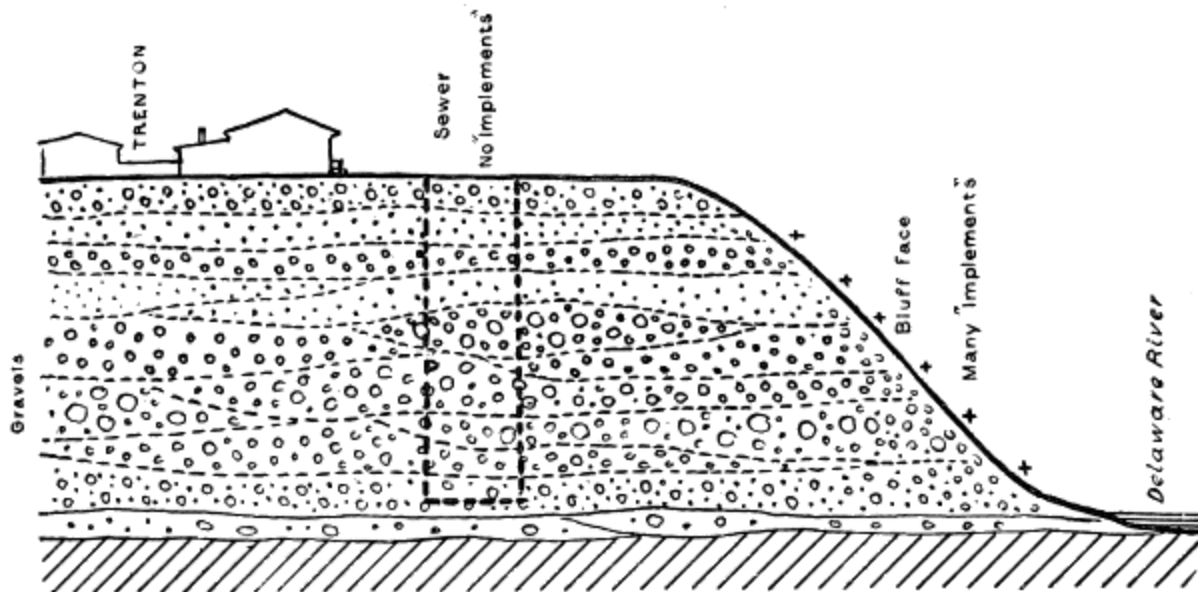


FIG. 2. Sections made by the river and by the sewer, the former yielding many "implements," the latter yielding none.

I made several visits to the place, descended frequently into the great cut and examined the gravels and their contents with the utmost care, but without securing a trace of art. Recognizing the vital importance of utilizing to the fullest extent this opportunity of testing the art-bearing nature of the gravels at this point, I resolved to undertake a systematic study of the subject. Summoning my assistant, Mr. William Dinwiddie, from his field of operations in the South, I had him spend upwards of a month at the great trench, faithfully watching the gravels as they were exposed. Mr. Dinwiddie had worked three years under my personal direction, and had helped open upwards of twenty trenches through similar gravel deposits, and was therefore well qualified for the work. Prof. W. J. McGee, Prof. R. D. Salisbury, Dr. Stewart Culin and Dr. Abbott also visited the place one or more times each. Relics of art were found upon the surface and in such portions of the talus as happened to be exposed, but nothing whatever was found in the gravels in place, and the search was closed when it became fully apparent that the case was hopeless.

It may be claimed that the conditions under which gravels are exposed in trenching as it progresses, are not as favorable for the collection of enclosed relics as where exposed by natural processes of weathering. This is true in a certain measure, as specimens may be obscured by the damp clinging sand which forms the matrix of the gravels. This, however, would

interfere but little with the discovery of large flaked stones, such as we were led to expect in this place, and this slight disadvantage in detecting shaped pieces in fresh exposures is more than over-balanced by the treachery of weathered surfaces which often give to intrusive objects the appearance of original inclusion. The opportunity for studying the gravels in all their phases of bedding, composition and contents, was really excellent, and no one could watch the constantly renewed exposures hour after hour for a month without forming a most decided notion as to the implement bearing qualities of the formation. Not the trace of a flaked stone, or of a flake or artificial fragment of any kind was found, and we closed the work with the firm conviction that the gravels exposed by this trench were absolutely barren of art. But Dr. Abbott claims to have found numerous implements in the bluff face a few feet away and in the same gravels. If this is true, the conditions of glacial occupation of this site must have been indeed remarkable. It is implied that during the whole period occupied by the melting of the ice sheet within the drainage of the Delaware valley the hypothetical rude race lived on a particular line or zone afterwards exposed by the river to the depth of 30 feet, leaving his strange "tools" there by the hundreds, while another line or zone, not more than forty feet away at most, exposed to the same depth by an artificial trench, was so avoided by him that it does not furnish the least memento of his presence. One vertical slice of the gravels twelve feet thick does not yield even a broken stone, while another slice not probably one-half as thick, cut obliquely through the gravels near by, has furnished subject-matter for numerous books and substantiation for a brace of theories. That no natural line of demarcation between the two section lines is possible, is shown by the fact that the formations are continuous, and that the deposits indicate a constant shifting of lines and areas of accumulation; thus it was impossible for any race to dwell continuously upon any spot, line or plane. This is well shown in the section, [Fig. 3](#), which gives the relations of the art-producing section of Dr. Abbott to the non-art-producing section of the sewer. The gravels were laid down entirely irrespective of subsequent cutting, natural or artificial; yet we are expected to believe that a so-called gravel man could have resorted for a thousand years to the space *a*, leaving his half shaped or incipient tools at all stages of the gravel building from base to top, failing entirely to visit a neighboring space *b*, or to leave there a single flake to reward the most faithful search. It is much easier to believe that one man should err than that

a guileless race should thus conspire with a heartless nature to accomplish such extraordinary results. The easier explanation of the whole matter is that the objects found by Dr. Abbott were not really in the gravels, but that they are Indian shop-refuse settled into the old talus deposits of the bluff, and that his eager eyes, blinded by a prevailing belief in a paleolithic man for all the world alike, failed to observe with their wonted keenness and power.

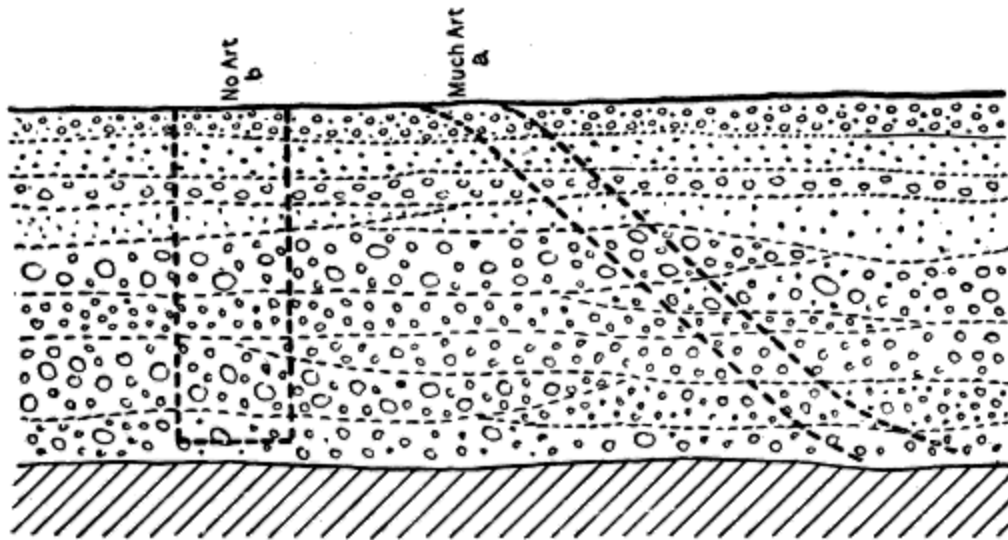


FIG. 3. *a*, Reputed "implement" producing zone of the river front. *b*, Barren zone of sewer.

But this case does not stand alone. The first discoveries of supposed gravel implements are said to have been made when the Pennsylvania Railway opened a road bed through the creek terrace on the site of the present station. At first numerous specimens of rudely flaked stones were reported, and the locality became widely known to archeologists, but the implement bearing portions of the gravels—and this is a most significant fact—were limited in extent, and the deposit was soon completely removed, the horizontal extension containing nothing. At present there are excellent exposures of the full thickness of the gravels at this point, but the most diligent search is vain, the only result of days of examination being a deep conviction that these gravels are and always were wholly barren of art.

It thus appears that here as well as upon the river front, the works of art were confined to local deposits, limited horizontally but not vertically, and a strong presumption is created that the finds were confined to redistributed

gravels settled upon the terrace face in the form of talus. Dr. Abbott states that "at that point where I gathered the majority of specimens there is a want of stratification."^[2] It is well known that such rearranged deposits are often difficult to distinguish from the original gravels. In trenching an implement producing terrace at Washington—where the conditions were probably quite similar to those at the Trenton railroad station—I passed through eighty feet of redistributed talus gravels before encountering the gravels in place, and so deceptively were portions of these deposits re-set that experts in gravel phenomena were unable to decide whether they were or were not portions of the original formation (cretaceous). The question was finally settled by the discovery of artificially shaped stones in and beneath the deposits.

^[2] Abbott, C. C. 10th Annual Report of the Peabody Museum, p. 41.

Again, an implement bearing deposit of gravel was recently discovered by the late Miss F. E. Babbitt at Little Falls, Minnesota, and sufficient (a very little) digging was done to satisfy the discoverer, and all paleolithic archeologists as well, that the objects were really imbedded in the glacial gravels. In the summer of 1892 I visited the place and carried a trench twenty feet horizontally into the terrace face on the "implement bed" level before encountering the gravels in place. The talus deposits were several feet thick, and were of such a nature that their true character could not be determined without careful and extensive trenching. The whole talus deposit was here well stocked with Indian quartz quarry-shop rejects, which were as usual of paleolithic types, and it was but natural that Miss Babbitt's conclusions, although based as they necessarily were upon inexperienced observations, backed by such well known "types" of "implements" should be unhesitatingly accepted by believers.

The occurrence of these telling examples of the deceptive appearance of re-set gravels would seem to justify and emphasize the conviction created by a critical examination of the two leading so-called paleolithic sites at Trenton, that Dr. Abbott, notwithstanding his asseverations to the contrary, has been deceived. Very strong support, it seems to me, is given to this conclusion by the recently published opinion of the late Dr. H. Carvill Lewis, a glacialist familiar with the Trenton region, and with the work of Dr. Abbott at the period of his paleolithic castle building. Dr. Lewis is

reported to have maintained before an open meeting of the Academy of Science in Philadelphia "that what Dr. Abbott believed to be undisturbed layers (of gravel) were those of an ancient talus."^[3] This remark may refer to both the main sites—the one at the railroad station and the other at the river front—or possibly only to the former. I have also heard it stated that that eminent scholar, Dr. Leidy, who must have had ample opportunities of forming correct opinions upon the subject, held pretty much the same views of Dr. Abbott's finds.

^[3] Brinton, D. G., *Science*, Oct. 28, p. 249.

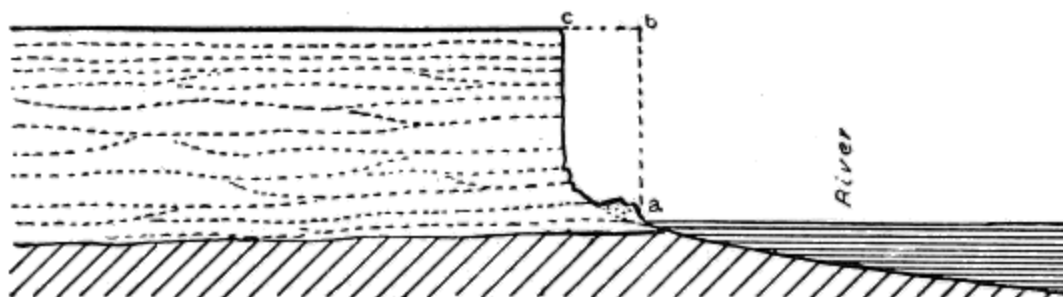


FIG. 4. A freshly formed gravel bluff.

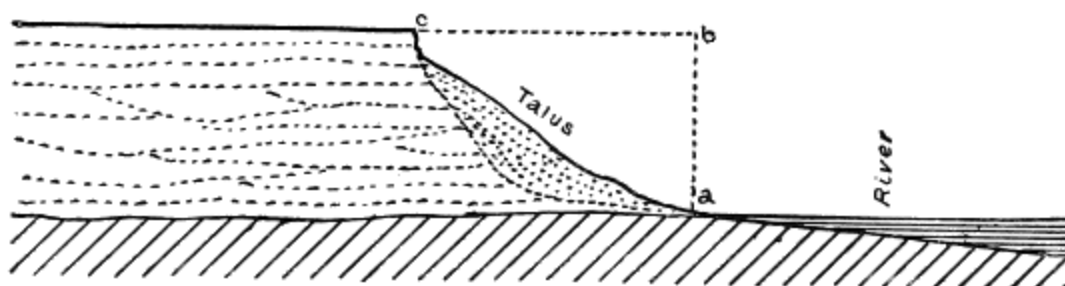


FIG. 5. Early stage of talus formation.

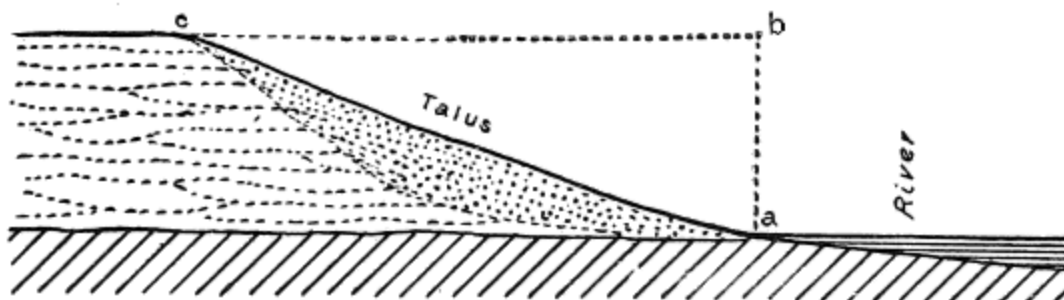


FIG. 6. An ancient talus.

To make the above criticism entirely clear, a few words of explanation of talus phenomena may be added. As a river cuts its channel deeper and

deeper into deposits of gravel a section is gradually exposed, but the gravels break down readily under atmospheric influences and the exposed face does not retain a high angle. The upper part crumbles and descends toward the base, there to rest against the slope or to be carried away by the stream. A supposititious case will be convenient for illustration. A gravel terrace twenty feet in height is encroached upon by the river at high water and undermined, and the face breaks down vertically, leaving an exposure as illustrated in [Fig. 4](#). In a very short time the upper portions become loosened and fall below, giving a steep slope as seen in [Fig. 5](#). The process goes on with gradually decreasing rapidity, and if the river does not again encroach seriously, a practically stable slope is reached, as shown in [Fig. 6](#). Such a talus may be hundreds or even thousands of years old, but there is rarely any means of determining its exact age. If the gravels are homogeneous in character, the talus will simulate their normal condition so completely that the distinction cannot be made out in ordinary gullies or by unsystematic digging. If the gravels contain varied strata the talus will be composite, and will be more readily distinguished from at least portions of the material in place.

Now it is important to observe what may be the possible art contents of such a talus as that shown in [Fig. 6](#). It may contain all objects of art originally included in that portion of the gravels represented by *a*, *b*, *c*, together with all articles that happened to be upon the surface *b*, *c*, beside such objects as may have accumulated from dwelling or shop work upon its own surface, after the slope became sufficiently reduced to be occupied for these purposes. A talus is therefore liable to contain, and in the utmost confusion, relics of all periods of occupation, supposing always that there were such periods, from the beginning of the formation of the gravel deposits down to the present moment. As a rule such a talus, if art-containing, will have a large percentage of shop and quarry-shop refuse, for the reason that the exposed gravels, and the banks and beds of rivers cutting them, furnish, as a rule, a good deal of the raw material utilized by workers in stone, and the shops in which the work was done are usually located upon the slopes and outer margins of the terraces. Although there is the possibility of very considerable age for these talus deposits, it is unlikely that any of them date back as far as the close of the glacial epoch or at all near it, for rivers change back and forth constantly, undermining first one

bank and then the other, so that a very large percentage of our talus deposits have been formed well within the historic period.

At Trenton the constantly exposed gravel banks afforded considerable argillite in boulders, fragments and heavy masses, as well as some other flakable stones of inferior quality little used, and it is inevitable that the Indian who dwelt upon the shores of the river should have sought the workable pieces along the bluff, leaving the refuse everywhere; and it is a necessary consequence that the terrace margin, the bluff face, and the talus deposits, places little fitted for habitation, should for long distances contain no trace of any art shapes save such as pertain to manufacture. Thus are fully and satisfactorily accounted for all the turtle backs and other rude forms that our paleolith hunters have been so assiduously gathering. Nothing can be more fully apparent than that no other race than the Indian in his historic character and condition need be conjured up to reasonably account for every phase and every article of the recovered art. Mistaken interpretations of the nature of shop rejects, and the common association of these objects with redistributed gravels, are probably accountable for the many misconceptions that have arisen. Talus deposits form exceedingly treacherous records for the would-be chronologist. They are the reef upon which more than one paleolithic adventurer has been wrecked.

Relics of art attributed to gravel man have been collected, so far as I can gather from museum labels and from incidental references in various publications, from a number of sites aside from the two already referred to. These are scattered over the city, and the finds were made mostly in exposures of the gravels that remained visible for a short time only, as in street and cellar excavations and well pits. These reported finds can never be brought within the range of re-examination, and the searcher after unimpeachable testimony must content himself with placing them in the doubtful column on general principles. Urban districts are so subject to disturbance through cutting down of hills, filling in of depressions, grading of streets, digging of foundations, cellars, sewers, wells and graves that no man can, from a limited exposure such as those producing the reported tools necessarily were, speak with certainty of the undisturbed nature of the deposits penetrated. It is doubtful if any one is justified in publishing such observations at all without serious query. Such testimony is liable to fall of its own inherent weakness, being absolutely valueless if unsupported by

collateral evidence of real weight. It can only be made permanently available to science by the discovery of something unusual or unique with which to couple it, something decidedly un-Indian in character or type, as for example the two skulls now in the Peabody Museum. These objects and the antler knife-handle exhibited with them may be alluded to as the only finds so far made at Trenton, having of themselves the least potentiality as proof and these skulls and this knife-handle must yet be subjected to the rigid examination made necessary by the importance of the conclusions to be based upon them.

Something may now be said concerning the art remains upon which this discussion hinges, and upon which conclusions of the greatest importance to anthropology are supposed to depend. Let us pass over all that has been said with regard to their manner of occurrence and association with the gravels and ask them simply what story they tell of themselves. Does this story, so far as we are able clearly to read it, speak of a great antiquity and a peculiar culture, or does it hint rather at vital weaknesses in the position taken by the advocates of these ideas? We shall see. The history of the utilization of rudely flaked stones in the attempt to establish a gravel man in America has never been written, but as read between the lines of paleolithic literature, it runs about as follows: The theory of a very rude and ancient people, having a unique culture and certain peculiar art limitations, was developed in Europe many years ago in a manner well known and often rehearsed. This people was associated with the ice age in Europe, and this epoch, with its moraines and till and sedimented gravels, was found to have been repeated in America. It was the most natural thing possible that these discoveries should carry with them the suggestion that man may have existed here as in Europe during that epoch, and that his culture was of closely corresponding grade. These were legitimate inferences and warranted the instituting of careful researches, but it was a dangerous suggestion to put into the minds of enthusiastic novices with fertile brains and ready pens. The idea was hardly transplanted to American soil before finds began to be made. The so-called "types" of European paleoliths suggested the lines upon which finds here should be made, and everything in the way of flaked stones connected directly or indirectly with the glacial gravels which had not yet been fully credited to and absorbed by the inconvenient Indian, was seized upon as representing the ancient time and

its hypothetical people and culture. In the early days of the investigation the various rude forms of flaked stones, resulting from failures in manufacture, had not been studied, and were shrouded in convenient mystery, and they thus became the foundation of the new archeologic dynasty in America, the dynasty of the turtle-back. Dr. Abbott states in his first work^[4] that these rude "implements" are not especially characteristic of any one locality, but seem to be scattered uniformly over the state. Specimens of every type, he says, are "found upon the surface, and are plowed up every spring and autumn; but this in no way militates against the opinion that these ruder forms are far older than the well-chipped jasper and beautifully-polished porphyry stone-work."^[5] At that stage of the investigation it was not at all necessary that a specimen should come from the gravels in place or from any given depth, since the "type" was supposed to be easily recognized and was a sufficient means of settling the question of age.

[4] Abbott, C. C. The Stone Age in N. J., Sm. Rep. 1875, p. 247.

[5] Ibid, p. 252.

Rude "implements" were called for and they were found. The only requirements were that they should not be of well-known Indian types, that they should be rude and have some sort of resemblance to what were known as paleolithic implements abroad. Since most of these so-called gravel implements of Europe are also doubtless the rejects of manufacture resemblances were readily found. The early attempts to utilize these rejects in support of the theory, and make them masquerade creditably as "implements" with specialized features and self-evident adaptation to definite ice-age uses, now appear decidedly amusing. Gradually, however, the lines have been drawn upon this early license, and it is to-day well understood by all careful students, that since the rude forms are so often repeated in modern neolithic refuse, the only reliable test of a gravel "implement" is its occurrence in the gravels in place. That a particular "implement," said to have been obtained from the gravels, is of "paleolithic type," does not in the least strengthen its claims to being a *bona fide* gravel implement; nor does its easy assignment to a "type" give any additional value to the collector's claim that the gravels said to contain it are implement bearing. The very names, "rude implement," "paleolithic implement," etc., carry with them a certain amount of mysterious suggestion; one thinks of unique, significant shapes and of strange, archaic

uses. At their mere mention, the great ice sheet looms up with startling realism, and the reindeer and the mighty mammoth appear upon the scene. The reader of our paleolithic literature is led to feel that these antiquated objects carry volumes of history in their worn and weather-beaten faces, but this is all the figment of fertile brains. These objects have without exception the appearance of the most commonplace every-day rejects of manufacture without specialization and without hidden meaning. They tell of themselves no story whatsoever, save that of the oft-repeated failure of the aboriginal blade maker in his struggle with refractory stones. This will be shown with greater clearness farther on.

But the scheme does not end with the repetition of a European state of affairs. Our gravel archeologists have not been content to adopt that feature of the foreign scheme which utterly destroys the paleolithic race before a higher culture is brought upon the scene. It was thought to improve upon the borrowed plan by allowing for a gradual development upward from the paleolithic stage, represented exclusively by a class of meaningless bits of flaked stone, through a period less rude, characterized by productions so far advanced as to be assigned to a definite use. These latter productions consist mainly of rather large and often rude blades, sometimes plain, but generally notched or modified at the broader end as if to be set in a handle, or attached to a spear or arrow shaft. These were assigned to post-glacial times in such a way as to bridge or partly bridge the great space between the glacial epoch and the present. They were separated arbitrarily from the body of the collections of the region, and referred to as probably the work of an Eskimo race. This arrangement produced a pleasing symmetry and completeness, and brought the history of man down to the beginning of the Indian epoch, which is represented by all of those forms of art with which the red man is historically associated.

Three principal periods are thus thought to be represented by the finds at Trenton; and in the arrangement of the collections these grand divisions are illustrated by three great groups of relics, which are looked upon by the founders of the scheme as an epitome of native American art and culture. By others this grouping is looked upon as purely empirical, as an arbitrary separation of the normal art remains of the historic Indian, not suggested by anything in the nature or condition of the objects, nor in the manner of their discovery.

The "Eskimo" feature of the scheme requires a more detailed examination than can be given it here. It may be stated, however, that the separation of the so-called Eskimo spear points, or whatever they may be, from the great body of associated articles of flaked stone, appears to be a highly arbitrary proceeding. That they were extensively made by the Indians is proved by the occurrence of refuse resulting from their manufacture on modern shop sites, and that they were used by the Indian, is equally apparent from their common occurrence on modern dwelling sites. The exceptionally large size of the argillite points is readily accounted for by the nature of the material. It was the only stone of the region well adapted to the manufacture of long blades or projectile points. Jasper, quartz and flint have such minute cleavage that, save in rare cases, small implements only could be made from them. Their peculiar manner of occurrence, described at so much length by Dr. Abbott,^[6] has been given undue consideration and weight. The phenomena observed may all be accounted for as a result of the vicissitudes of aboriginal life and occupation within the last few hundred years as fully and satisfactorily as by jumping thousands of years backward into the unknown.

[6] Abbott, C. C. Popular Science Monthly, Dec., 1889.

Whatsoever real support there may be for the "Eskimo" theory, either in the published or the unpublished evidence, it is apparent that under the present system of solitary and inexpert research, the scientific world will gain little that it can utilize without distrust and danger. Whatsoever may be the final outcome—which outcome is bound to be the truth—it is clear that there is little in the present evidence to warrant the separation of a "paleolithic" and an "Eskimo" period of art from that of the Indian.

That the art remains of the Trenton region are essentially a unit, having no natural separation into time, culture or stock groups, is easily susceptible of demonstration. I have already presented strong reasons for concluding that all the finds upon the Trenton sites are from the surface or from recent deposits, and that all may reasonably be assigned to the Indian. A find has recently been made which furnishes full and decisive evidence upon this point. At Point Pleasant, on the Delaware, some twenty-five miles above Trenton, there are outcrops of argillite, and here have been discovered recently the shop sites upon which this stone was worked. There are two

features of these shops to which the closest attention must be given. The first is that they are manifestly modern; they are situated on the present flood plain of the Delaware, and but a few feet above average water level, the glacial terrace here being some forty or fifty feet in height. These shops, therefore, represent the most modern phases of aboriginal industry, and may have been occupied at the coming of William Penn. The second point is that every type of flaked argillite found in the Trenton region, associated with the gravels or otherwise, is found on this site. It was to a certain extent a quarry site, for the great masses of argillite brought down by the floods were here broken up and removed from the river banks or bed. It was a shop site, for here the articles, mainly blades, were roughed out, and it was also a dwelling place—a village site—where all the specialized forms of flaked stones made from the blades were prepared for use. Here are found great numbers of the rude failures, duplicating every feature of the mysterious "paleolith" with which our museums are stocked, and exhibiting the same masterly quitting at just the point "where no further shaping was possible."^[7] Here we see the same boldly manipulated "cutting edge," the "flat bottom" and "high peak," and the same mysteriously weathered and disintegrated surfaces, so skillfully made, by a nice balancing of accidents,^[8] to tell the story of chronologic sequence in deposition.

^[7] Abbott, C. C. Smithsonian Report, 1875, p. 248.

^[8] Ibid. Primitive Industry, p. 487.

Beside the failures, we have here, as on other quarry shop sites, the evidence of more advanced work, the wide, thick, defective blades, and many of the long, thin blades broken at or near the finishing point. Here, too, just back of the roughing-out shops, are the dwelling sites from which many specialized forms are obtained. The "Eskimo" type is fully represented as well as the ordinary spear point, the arrow point, and the perforator of our Indian. There is not a type of flaked argillite known in the Delaware valley that may not be duplicated here on this modern Indian site, and this has been known by local archeologists for years. Why so little has been said about the matter is thus explained. Dr. Abbott, in 1890, discovering this site, and finding "typical paleolithic implements" (the ordinary ruder forms of rejects) among the refuse, was so entirely at a loss to explain the occurrence that he felt compelled to again "take up the examination of the gravel deposits of the valley of the Delaware" with the

hope of "finally solving the problem."^[9] The true conditions would have been at once apparent to any one not utterly blinded by the prevailing misconceptions.

[9] Abbott, C. C. Annual Report of the Curator of the Museum of American Archeology, University of Pennsylvania. No. 1, p. 7.

The entire simplicity of the archeologic conditions in the Delaware valley may be further illustrated. Had William Penn paused in his arduous traffic with the tawny Delawares, and glanced out with far-sighted eyes from beneath the pendant branches of the great elm at Shackamaxon, he might have beheld an uncouth savage laboriously fabricating rude ice age tools, making the clumsy turtle-back, shaping the mysterious paleolith, thus taking that first and most interesting theoretical step in human art and history. Had he looked again a few moments later he might have beheld the same tawny individual deeply absorbed in the task of trimming a long rude spear point of "Eskimo" type from the refractory argillite. If he had again paused when another handful of baubles had been judiciously exchanged, he would have seen the familiar redskin carefully finishing his arrow points and fitting them to their shafts preparatory to a hunting and fishing cruise on the placid Delaware. Thus in a brief space of time Penn might have gleaned the story of the ages—the history of the turtle-back, the long spear point and their allies—as in a single sheaf. But the opportunity was wasted, and the heaps of flinty refuse left upon the river bank by the workmen were the only record left of the nature of the work of that day. Two hundred years of aboriginal misfortune and Quaker inattention and neglect have resulted in so mixing up the simple evidence of a day's work, that it has taken twenty-five years to collect the scattered fragments, to sift, separate and classify them, and to assign them to theoretic places in a scheme of culture evolution that spans ten thousand years.

Yet is there really nothing in it all, in the theories, the observations, the collections and the books? Do I speak too positively in condemnation of the results of years of earnest investigation? Perhaps so, but the voluminous testimony is so overloaded with inaccuracies, the relics of unscientific method and misleading hypotheses, that every item must be sharply questioned; and the conclusions reached so far overstep the limits warranted by the evidence, that heroic measures alone can be effectual in determining their exact value. If, as many believe, vital errors have been embodied in

the evidence presented by the advocates of the theory, it is impossible to state the case too strongly. Error once fully absorbed into the literature of science has many advantages over the tardy truth; it is strongly fortified and must be attacked and exposed without fear or favor. Truth involved with it cannot permanently suffer. If the twin theories of a gravel and a paleolithic man in eastern America are to be assailed as unsound or as not properly supported, it should be done now while the originators and upholders are alive and alert to sustain their positions or to yield to the advances of truth. I do not wish to wrongly characterize or to unduly minimize the evidence brought to bear in favor of these theories. I do intend, however, to assist the world so far as possible in securing an exact estimate of all that has been said and done, and all that is to be done.

In a previous article I have examined the evidence relating to paleolithic art in the eastern United States, and have indicated its utter inadequacy and unreliability. In this paper the testimony relating to the occurrence of gravel art, in the locality most fully relied upon by advocates of the theory, has been partially reviewed and subjected to the strong light of recent observations. It is found that the whole fabric, so imposing in books and museums, shrinks away surprisingly as it is approached. The evidence furnished by the bluff face and by the railway cutting, the two leading sites, is fatally weakened by the practical demonstration of the fact that the gravels proper are at these points barren of art remains. In endeavoring to naturalize an immigrant hypothesis, our gravel searchers, unacquainted with the true nature of the objects collected and discussed, and little skilled in the observation of the phenomena by means of which all questions of age must be determined, have undoubtedly made grievous mistakes and have thus misled an expectant and credulous public.

The articles themselves, the so-called gravel finds, when closely studied are found to tell their own story much more fully and accurately than it has heretofore been read by students of archeology. This story is that the art of the Delaware valley is to all intents and purposes a unit, that there is nothing unique or especially primitive or ancient and nothing un-Indian in it all. All forms are found on demonstrably recent sites of manufacture. The rude forms assigned by some to glacial times are all apparently "wasters" of Indian manufacture. The large blades of "Eskimo" type are only the larger blades, knives and spear points of the Indian, separated arbitrarily from the

body of the art-remains to subserve the ends of a theory, certain obscure phenomena of occurrence having been found to give color to the proceeding. To place any part of this art, rude or elaborate, permanently in any other than the ordinary Indian category will take stronger proofs than have yet been developed in the region itself.

The question asked in the beginning, "Are there traces of glacial man in the Trenton gravels?" if not answered decisively in the negative, stands little chance, considering present evidence, of being answered in the affirmative. In view of the fact that numerous observations of apparent value have been made in other sections, there is yet sufficient reason for letting the query stand, and we may continue to cherish the hope that possibly by renewed effort and improved methods of investigation, something may yet be found in the Trenton gravels clearly demonstrative of the fascinating belief in a great antiquity for the human race in America.

The evidence upon which *paleolithic man* in America depends is so intangible that, unsupported by supposed analogies with European conditions and phenomena, and by the suggestions of an ideal scheme of culture progress, it would vanish in thin air; and if the theory of a *glacial man* can summon to its aid no better testimony than that furnished by the examples examined in this paper, the whole scheme, so elaborately mounted and so confidently proclaimed, is in imminent danger of early collapse.

W. H. HOLMES.

GEOLOGY AS A PART OF A COLLEGE CURRICULUM.

The demand for scientific studies as a part of the college curriculum is felt by all those who have to do with the provision of higher instruction for American youth. The reasons for this may be various, but a fundamental reason is found in the tendency among the American people in particular, and in this age in general, toward practicality in all things. Applied to education this practicality asks for a training which shall have a direct bearing upon the business of life to be followed immediately after the training period is ended. It means a differentiation of subjects and specialization in methods to adjust the education to the different functions which the students taking it are preparing for. It calls for a professional education for those who expect to become lawyers, doctors, ministers, or teachers,—a technical education for those who are to engage in the arts of the mechanical or civil engineer, or of the architect. It results not only in the establishment of colleges and universities devoted to this kind of education, but it affects the methods of the high schools and academies, and is felt down to primary schools, and on the other hand the older institutions founded on a different plan are adapted to the popular demand by the addition to the regular studies of "electives," chosen not always for their value or disciplinary studies, but because of the practical applicability of the information to be derived from them, to the business of the student.

Without discussing the relative merits of the two ideas of education, the chief contrast between them may be found in the character of the results sought. The knowledge of things and their uses is of chief importance in the practical education; the knowledge of ideas and skill in their use is the aim of the liberal education. Geology is one of the sciences which most men will at once classify as among the practical sciences. It deals with matters of practical importance to everybody. Coal, iron, the metals, silver, gold, tin, lead, building stone, sand, clay, petroleum, and natural gas, and all geological products are essential materials of modern civilization, and a knowledge of them and of their modes and places of occurrence is one of

the requisites of an education, either from the practical or the liberal point of view. So too the dynamics of atmospheric and hydraulic erosion, the agency of rivers and oceans in destruction, removal and reconstruction of geological formations have their eminently practical bearings upon the various arts of engineering. While the practical value of geology is thus evident and undisputed, it is not on this account that its importance as a part of a college course of education is urged. As a practical study geology becomes the centre of a group of studies requiring years for mastery. Chemistry and physics are primarily essential to a full understanding of the most common of geological problems. And to use geological facts and phenomena, an acquaintance with the complex methods of engineering, civil and mechanical, which again call for a thorough mastery of mathematics, is necessary. Mineralogy and petrography, metallurgy and mining engineering have each reached a stage of development entitling them to the rank of separate sciences, but the practical training of the geologist should include them all. When we add the biological sciences connected with historical geology, paleontology, zoölogy and botany, with all the laboratory and field work required for their proper study, we have a group of affiliated branches of learning requiring four or five years of continuous study after the student has learned how to study. It is plain therefore that only a specialist, one who is willing to neglect other studies, or who has previously had a liberal training, can perfect himself on the practical side in the science of geology.

But irrespective of its practical uses, as a means of training and supplementary to the ordinary studies of a college curriculum, geology is one of the most useful of the sciences of observation. It is in providing that particular training to which President Eliot has recently called attention in the *Forum* (Dec., 1892, *Wherein Popular Education Has Failed*), that geology can be used to such advantage. Speaking particularly of the lower education, President Eliot says it is "the judgment and reasoning powers" that particularly require attention. Their systematic development is to be attained in the four directions of "observing accurately, recording correctly, comparing, grouping and inferring justly, and expressing cogently the results of these mental operations." (p. 421.) The attainment of these ends is one of the purposes of liberal education, whether it be in the primary school or in the university. And geology, or any other science, is of value in a

college course in proportion to its fitness for the exercise and development of these functions of the student. Geology may be taught without regard to these ends, and then it is valuable from the practical point of view, but when we examine it in respect of its availability as a disciplinary study we find it offering particular attractions.

Using the distinction between theory and practice, which is as old as Aristotle, geology in its theoretical aspect is more easily comprehended than is the theoretical aspect of most of the modern sciences. This arises first from the fact that the facts and phenomena are of a simple and grand nature, making it possible for the teacher to direct certain attention to the specific facts under consideration. The water of the rivers, the mud by the road side, the rocks and sands on the shore are familiar objects to all, and it is a simple matter to call attention by ordinary language to the specific facts regarding them, which, analyzed out, are to form the basis of exact ideas and scientific definition and classification. Geology is the one science among the natural sciences which may begin with the common language of the pupil, and by means of such language alone may build up ideas of precise phenomena in scientific terms. Physiography or physical geography surpasses geology proper in this particular, as the admirable work of Professor Davis is showing, and on this account it is the best introduction to geology. But the very largeness and indefiniteness of the facts are in the way of the use of physical geography for the exercise of the finer and more exact functions of observation. The disciplinary value of classics and mathematics is to a considerable extent derived from this quality, the precision with which the words or figures kindle like ideas. So long as the object of the training is to teach the knowledge of ideas and how to use them, classics and mathematics are the simplest and purest means of developing a liberal education. The addition of sciences to the college course is not because of the usefulness of the knowledge of things thus to be gained, but because the language of the sciences is essential to call forth the observation and the exercise of the accompanying mental operations.

When it comes to dealing with the ideas associated with particular sense-observation, where form or motion can not be expressed in simple mathematical terms, language can not communicate a new idea or kindle it in another mind with precision. It is necessary by some means to recall or to present the object itself to the student. In the teaching of science this point

is of great importance, and much of the unsatisfactoriness of science-teaching is doubtless due to failure to note it. No circumlocution of words can arouse in another or communicate to him the idea appropriate to a sensation he has never felt. The blind man whose eyes are opened sees men as trees walking.

In the use of science for elementary training (and the training is elementary until the student is capable of investigating and interpreting the facts and phenomena of a science directly) that science is the better which deals with objects which are simple, common and easily observed. Such is geology in some of its aspects. Every time the student walks in the country he sees the facts discussed in the text-book or by his teacher; and from attention to those with which he is already familiar he can be readily led to observe and give attention to others and to analyze those already in his mind by properly directed questions.

In the field of geology are found the ready means for the exercise and development of observation and thought. The learner begins with ideas which every intelligent mind associates with the objects described or named, and by degrees the marks of his knowledge are increased, the relations of things are grasped, and the content of his ideas associated with the language of his science is enlarged. In the process of learning the science he has been building up his stock of knowledge of facts and phenomena, but, of more importance than that, he has learned the method of observing and of scientific thinking. He has had training in the methods of reducing the hard facts of nature to the laws of thought and practice, he has seen the method by which theoretical order is made out of the interminable confusion and complexity of natural things.

Beside this primary reason for the use of geology as a disciplinary science-study, there is a second reason arising from the symbolic nature of a large group of its facts. This aspect of the science is best seen in the historical and stratigraphical parts of geology, in which fossils are the chief data for study. The interpretation of a fossil into a species of organism, having its definite place in the elaborate classification of the zoölogist, or as an indicator of the time and place and mode of formation of the strata in which it is buried, is, to be sure, a most intricate and, at first thought it would seem, an unattractive process. But no more so, I would say, than the

interpretation of a series of Greek characters. The interpretation of the Greek reveals to us the richest results of human thought and most perfect laws of human speech, and we find therefore in the analysis required the most perfect discipline of the powers of speech and language. The fossil too holds, ready to be revealed, the story of the history of the world and the laws of the evolution of the organic life of the globe, and records an inexhaustible wealth of information regarding the laws of nature. But as an instrument of intellectual discipline its great merit lies in its symbolic nature. It is this symbolic character of the classical languages and of the mathematics which fits them to be universal means of liberal training. The symbolic nature of the fossil fits it to become the exponent of training in the pure science of nature. The fossil is a mark which stands for something, and thus, in the nature of things, it asks for interpretation. As a symbol it stimulates minute and accurate observation, and kindles close and exhaustive thought; as a symbol it leaves us the ideas it has engendered after it is lost to memory as an observation. Thus the value of its study does not depend upon the retention in the memory of the facts brought before the mind, but in the training of the mental processes required in its interpretation. The study of this branch of geology exercises and develops all the faculties which are specially exercised in any scientific investigation.

Another aspect in which it is an ideal means for such training comes from the fact that it is equally valuable at every stage of progress of the student. When first examined it means nothing to him. He knows nothing of organism, of strata, of geological time. The fossil gains meaning only as he is able to put meaning into it. The student must ask questions, and as step by step he answers his questions by more minute and wider examination, the fossil holds a fuller interpretation. His studies lead him to investigation of the whole field of nature, the rocks, the formation of deposits, the action of the elements, the conditions of life, the forms of organism, their functions and habits, the laws of growth, their adaptation to environment, the changes of events in time, the efforts of association and struggle for life, the principles of evolution and development—the migration and origin and extinction of organisms on the globe. Nothing in nature is without interest to him. Further than this the amount of good he gains is not measured by the number of fossils he studies, but by the wideness of his research. A handful of fossils from some one fossiliferous ledge may be the text for a

year's study, and the methods acquired in the study may be the nucleus of a life's work. In this department of geology the possibilities for new discoveries, new developments of science are almost endless. As a single author thoroughly read develops a wealth of knowledge of the laws of language and thought, so geology may be studied by the use of a limited set of its phenomena and become the introduction to the exhaustive study of natural science.

Another advantage attaching to geology as a science-study for the college curriculum, arises from the fact that it may be pursued deeply without the elaborate aid to the senses required in other sciences for making minute record or measurement of facts or phenomena. As in language and mathematics, it is essential to acquire a familiarity with the grammar, the dictionary and the symbols, formulas and rules of their usage before the finer training in the use of thought begins, so the vocabulary and the definitions of a science must be acquired before much use can be made of the higher discipline to be derived from scientific study. In language study this higher training comes from practice in making the minute analysis, in detecting the fine shades of meaning expressed in the literature itself. So it is important in selecting a science to be used as a disciplinary study that the facts and laws of nature with which it is concerned should be capable of clear and precise definition, and, moreover, that it should furnish a field for the study of the minute and intricate relationship existing between the different facts which are to be attained by personal inspection of the objects themselves. In most of the sciences this deeper exercise of scientific thought requires for its successful pursuit artificial aids to the common senses of observation. Chemistry must have its purified acids and reagents, test tubes, and delicate scales for measurement of weight and volume. Mineralogy must have its chemical analyses, or optical measurements so fine that microscopes of highest power are essential tools for the investigation. Physics must have the most delicate measurements of time and space and weight. Botany, for the earlier stages of study, is fully equal to geology in these respects, but its scope is much less general. Zoölogy requires dissections calling for skill in manipulation, and in other respects is ill adapted to general classes. But precision in the intellectual processes of observation and reasoning can be cultivated in the use of geological facts to their highest and widest perfection, with scarcely any aids to the normal

faculties of observation. A couple of hammers, a pocket lens, a chisel and a few pointed steel tools for revealing fossils, a tape line, compass and clinometer are the few equipments that will enable the geologist to carry his investigations to almost any degree of thoroughness.

What has already been said applies to the study of the pure science of geology either in the field or in the laboratory. There is still another use to which this, as other sciences, may be put in disciplining the college student in directions not provided for by literary or mathematical studies,—the study of man as an investigator. In the pursuit of the study of geology, the first instruction must be received in didactic form, but after the text-book and lecture stage is passed, or while it is under way consultation of the literature of the sciences is appropriate. In the use of scientific literature the critical judgment is brought under training, and the varying interpretations of well known phenomena by expert scientists suggest the prominent part which the notions already in the mind play in the interpretation of the external facts observed. The experienced geologist will recall many cases of honest report of impossible facts by men who are unable to distinguish between what they saw and the false interpretations they made of these observations. One man will report that a live toad jumped out of the middle of a solid piece of coal, when it was heated in the stove; another will swear that he saw a fossil shark's tooth taken out of a ledge of Trenton limestone. It is evident that our memory of observation is not the revival of the object producing the sensation, but of the idea we framed of the sensation at the time. The study of original descriptions of objects of nature reveals the fact that the describer uses the ideas he already has in his mind as he does the standard foot-rule in his hand for measuring that which he describes, and it is by the study of scientific literature and the comparison of views of many scientists that this highest discipline of the observational faculties is attained—the power to determine the personal equation of error for the observer, and thus see through his descriptions a truer representation of the facts than the observer himself saw. Geological literature is admirably adapted for this higher discipline, and in no field of science (I think not in astronomy itself), has wider and more comprehensive thought been applied than in geology. While other branches of science have been developed and become more narrow and special in their treatment of the facts concerned, geology still stands as the most comprehensive of all the sciences of nature.

H. S. Williams.

YALE COLLEGE, November 30, 1892.

THE NATURE OF THE ENGLACIAL DRIFT OF THE MISSISSIPPI BASIN.

It is of some importance, both to the practical work of the field and the theoretical deductions of the study, to determine the nature and amount of the drift that was carried forward in the body of the ancient continental glaciers, and brought out on their terminal slopes and at length deposited at their frontal edges, and to distinguish it from that which was pushed or dragged or rolled along at the bottom of the ice.^[10] It may be helpful to indulge in a speculative discussion at the outset to prepare the way for the specific evidence and the inferences to which it leads.

^[10] Debris, which may be imbedded in the basal layer of the ice during some part of its transportation, but which is brought down to the bottom and subjected to basal action in the latter part of its course, and ultimately becomes a part of the basal deposit, is not here included in the englacial drift.

Whenever a prominence of rock is overridden and enveloped by a glacier of the free-moving continental type, one of two things takes place; either that part of the ice which passes over the summit of the prominence flows down its lee slope, carrying whatever debris it dislodges down to the rear base, and thence onward along the bottom of the ice, or else the currents which pass on either side of the prominence close in behind it before the corresponding current which passes over the summit reaches the point of their junction, in which case the summit current is forced to pass off more nearly horizontally into the body of the ice, carrying with it whatsoever debris it has dislodged from the summit of the prominence and embodied within its base. The law of the phenomena appears to be that whenever the height of the prominence is less than one-half the base, measured transversely to the movement of the ice, the summit current will follow down the lee slope; but whenever the height of the prominence is more than one-half the transverse base, the lateral currents will close in on the lee, and the summit current will flow off into the body of the ice. This simple law is, however, subject to very considerable modifications from several different sources which may be grouped under (1) differences in the

friction arising from basal contact, and (2) differences of internal friction and mobility. The lateral currents will expose more surface to the sides and base of the hill and the adjoining plain, and will be more subject to conflicting currents, while, on the other hand, being deeper currents, they will presumably be more fluent. These and other qualifying conditions will go far to vitiate the application of the law, but its statement may have some value as representing a general conception of the phenomena. When the height of the prominence becomes great relative to the total thickness of the ice, the fluency of the summit current may be much reduced relative to that of the central parts of the lateral currents. When the prominence reaches the surface, blocks dislodged from it are borne away on the surface of the glacier, and constitute superglacial drift. Blocks dislodged from near the summit, but below the surface of the ice, are presumably carried onward in the upper zone of the glacier; while other blocks detached at various but sufficient heights on the side of the prominence are doubtless borne around into the lee and carried forward in the same vertical plane as the summit stream, so that there comes to be a vertical zone set with boulders moving on from the lee side of the nunatak.

Lofty ledges or plateaus, with vertical or undercut faces, furnish similar means for the lodgment of debris within the body of the ice.

In these and doubtless in other ways it appears that there came to be lodged directly within the body of the Pleistocene glaciers at some considerable distances above their bases, blocks derived from rock prominences that rose with sufficient steepness above the general surface of the country over which the ice passed. The lodgment of debris on the lateral borders of glaciers is neglected here because it has little or no applicability to the phenomena of the upper Mississippi basin. It is also doubtful whether any prominences protruded through the ice except near the thin edge, when advancing and retreating, and these are too inconsiderable to merit attention.

It is obvious, upon consideration, that blocks detached from summits or from the sharp angles of out-jutting ledges or plateaus might suffer some glacial abrasion in the process of their dislodgment and transposition along the crest or projecting angle, but that in general such abrasion would be small, and, in most cases, nearly or quite absent. The debris so incorporated

in the body of the ice would be, for the most part, angular, and, as it was brought forward in the ice, it would probably suffer very little abrasion. If it continued to move forward in the plane in which it started, descending only so much as the bottom wastage of the ice required, it would be brought out to the terminal slope of the ice sheet by virtue of the melting away of the ice above, and thence it would be carried on down the terminal slope as superglacial debris, and dropped at the frontal edge. If this be the true and full history, there would be no commingling of this englacial matter with the subglacial debris. It is evident, that the englacial matter brought forward from the crest of one prominence would be intermingled with that brought forward from other prominences lying in a line with it, or lying so near it that the lateral spreading of the debris would lead to commingling. It is also clear that variations in the direction of currents would tend to the same result, so that englacial matter from different prominences of the same general region might be commingled. So also englacial material, by crevassing and by the descent of streams from the surface to the base, would be carried down to the bottom and mingled with the subglacial debris. So also blocks broken away from the base of the prominence which yielded the englacial erratics might be moved forward along the bottom parallel with the englacial material above, and lodged at any point along the line. It is therefore to be expected that the basal deposits will contain the same rock species as the englacial, but if there be no process by which the basal material is carried upward the reverse will not be the case, and there will be a clear distinction between the englacial deposit and the subglacial deposit, in composition as well as physical state.

Not a few glacialists, however, advocate in somewhat differing forms and phases the doctrine that basal material is carried upward into the body of the glacier and at length reaches the surface, and that at the extremity of the ice this is commingled with any erratics that may be englacial or superglacial by original derivation. This doctrine appears to have had its origin in the endeavor to explain the very common fact that glacial drift has been carried from lower to higher altitudes. Erratics are often found lodged several hundred feet higher than the outcrop from which they were derived. It has never seemed to me, however, that this phenomenon necessarily was different in kind from that which takes place in the bottom of every stream; at least I have not come in contact with any instances that seemed to require

a different explanation, except those connected with kames and eskers that require a special explanation in any case. We are so accustomed to view streams from above, and so accustomed to study the extinct glaciers from the bottom, that we are liable to overlook the community of some of the simpler processes involved alike in both phenomena. The dictum that water never runs up hill is measurably true of the surface currents of the ice as well as water, but it altogether fails when applied to the basal currents of either. It is probable that there is no natural stream of any length in which, at some part of its course, basal debris is not carried from lower to higher altitudes and lodged there. If the bed of any stream were made dry and the debris in it critically examined, it would be found that at numerous points the silts or sands or gravels had been carried from the bottom of some basin in its bed to the higher rim or bar or reef that bordered it on the downstream side. So I conceive that, on a grander scale, the natural result of the flow of the basal ice of a continental glacier over the inequalities of the country was the lifting of material from some of the lower horizons and its lodgment on the crests of ridges or the slopes or summits of mountains that lay athwart its course.

So again, it is certain that a considerable part of the peripheral drainage of glaciers takes place through tunnels beneath the ice. It is reasonable to suppose that during the winter season, when the drainage is slack, these tunnels tend to collapse in greater or less degree, under the continued pressure of the ice and the "fattening" of the glacier, so that in the early part of the next melting season the contracted tunnels may be over-flooded by glacial waters. To the extent that these tunnels become incompetent the water would become ponded back in the crevasses and moulins by which the surface-water gains access to them. They thus come to have something of the force of water flowing in tubes, and may be presumed to be capable of forcing rounded material to some considerable height, and of carrying ice-imbedded boulders to any point reached by the stream. These tunnels probably undulate with the bottom, and lodgment along them takes place wherever enlargement permits.

Without, therefore, appealing to any upward cross currents within the ice itself, it is possible to explain the transportation of the drift from lower to higher altitudes. I have never seen phenomena of this kind that seemed to call for any other explanation than these. I am not prepared to say that there

are no such phenomena. One of the purposes of this article will have been accomplished, if it shall call forth a critical statement of phenomena that require the assumption of internal upward movements of the ice to account for them, and of the criteria which distinguish such phenomena from those that may be referred to upward basal movements such as are common to all streams or to the exceptionally conditioned subglacial streams. That there are upward internal movements in most streams is as much beyond question as the existence of upward basal currents in rivers and glaciers, but they are dependent chiefly upon the velocity of the current and the irregularity of the bottom. Theoretically, as I understand, a stream moving in a straight course on a perfectly smooth bottom would not develop an upward cross current. Each lower layer would move slower than that above it by reason of basal friction, but they would move on in parallel lines. But if irregularity of bottom be introduced the parallelism is obviously destroyed, and if the velocity be high so that the momentum of the particles becomes great relative to their cohesion, irregular internal movements will result, and these will often be of a rotary nature in vertical planes bringing the basal parts of the fluid to the surface or the reverse. For this reason rapid streams abound in rotary currents, while slow streams do not.

Now it is quite obvious that a stream of water moving at a rate of three or four feet per day, or even fifty or sixty feet per day, would not develop perceptible upward currents, and certainly would not lift the lightest silt from its bottom. I do not think there are any theoretical grounds for believing that internal glacial currents are developed, which flow from base to surface, carrying bottom debris to the top.

One of the most remarkable expressions of the drift phenomena of the Upper Mississippi region consists of belts of boulders stretching for great distances over the face of the country, and disposing themselves in great loops after the fashion of the terminal moraines of the region with which they are intimately connected. Besides this, there are numerous patches of boulders of more or less irregular form and uncertain relations. The whole of these have not been studied in detail, but a sufficient portion of them have received careful examination to justify the drawing of certain conclusions from them. Those which have been most studied lie in Ohio, Indiana, Illinois, Michigan, Wisconsin, Iowa and Dakota. Those of the first three States have been most carefully traced and their constitution is such as

to give them the greatest discriminative value. To these our discussion will be limited chiefly.^[11]

[11] Parts of these tracts were long since described by Bradley of the Illinois Survey. (Geol. Surv. Ill., Vol. IV. p. 227). Collet of the Indiana Survey (An. Rep. 1875, p. 404) and Orton and Hussey of the Ohio Survey (Geol. Surv. Ohio, Vol. III., pp. 412, 414 and 475). The relationship of these tracts to morainic lines and to each other I worked out some years since (Third An. Rep. U. S. G. S. pp. 331, 332, 334) but I owe many details and some important additions to my associate, Mr. Leverett.

Emerging from the dunes at a point north of the Iroquois river in Jasper county, northwestern Indiana, a well characterized belt of surface boulders stretches westward to the State line, just beyond which it curves about to the south and then to the east, and re-enters Indiana a little south of the northwest corner of Benton county. It soon turns abruptly to the south and reaches the Wabash river near the centre of Warren county. The immediate valley of the Wabash is thickly strewn with boulders from the point where the belt reaches it to the vicinity of West Point on the western line of Tippecanoe county. The uplands, however, do not give any clear indication of the continuity of the belt, and the connection is not altogether certain. There is an inner well-marked belt that branches away from this in the central part of Benton county and runs southeasterly into the northwestern quarter of Tippecanoe county, beyond which only scattered boulders occur, which leaves its precise connections also in doubt. But starting from West Point, which is less than a dozen miles from the point where the two belts cease to be traceable with certainty, a well-defined belt, one or two miles wide, runs southeasterly across the southwestern corner of Tippecanoe county and the northeastern quarter of Montgomery county to the vicinity of Darlington, beyond which its connection is again obscure, although boulders occur frequently between this point and the northwestern corner of Brown county, where boulders are very abundant. So also, patches of exceptionally abundant boulders occur in the west central part of Clinton county. These may be entitled to be regarded as a connecting link between the train which enters northwestern Tippecanoe county and that of northwestern Boone county, as scattered boulders of the surface type, but of not very exceptionally frequent occurrence, lie between them. However this may be, a belt of much more than usually frequent surface boulders stretches southeasterly to the vicinity of Indianapolis, and probably

connects with a very well-marked belt lying near the south line of the southeast quarter of Marion county and in the northeastern part of Johnson county. There is also a well-defined tract in southeastern Hendricks county, running east and west, without evident connection with the foregoing tracts, though it may be the equivalent of the Darlington belt. There is also a somewhat unusual aggregation in the form of irregular belts in southeastern Johnson county, in the vicinity of Nineveh, and in southern Shelby county. The belt south of Indianapolis is probably to be correlated by scattered boulders only slightly more abundant than those of the adjacent region, but of the surface type, stretching northeasterly to near the center of the west half of Henry county, where a well-marked belt again sets in. From this point the tract runs northeasterly nearly to the north limit of the county, where it turns easterly and runs in the vicinity of the line between Randolph and Wayne counties to near the Ohio line, where it curves to the southeast entering Ohio near the northwest corner of Preble county. In its southeasterly course across that county it is phenomenally developed as has been well shown by the descriptions of Professor Orton. Soon after entering Montgomery county it curves about to a northeasterly course, and crossing the great Miami river, a few miles above Dayton, holds its northeast course across the southeastern part of Miami county, the northwestern part of Champaign county, and thence on to about the center of Logan county, where it curves about and runs in a direction a little east of south to near the southeast corner of Champaign county, beyond which it ceases to be a specially notable phenomenon.

In the region between the Wabash and Kankakee rivers, in northern Indiana, there are numerous tracts of irregular form over which surface boulders in phenomenal abundance are scattered. These are particularly noticeable in southern Jasper county; in the vicinity of Wolcott, Monon and Chalmers in White county; near Star City in Pulaski county; in the southeastern corner of Stark county, and very generally along the great interlobate moraines, lying parallel with the Eel river, and some others of the Saginaw glacial lobe. These are so associated with the inter-tangled morainic phenomena of that region as not to admit of convenient and brief description in their genetic relationships.

The well-defined tracts have a most significant distribution. The first part described is associated with the terminal moraine that marked the

margin of a lobe of ice that moved westward along the axis of the Iroquois basin to a point a few miles beyond the Indiana-Illinois line. The portion that runs southward to the Wabash is associated with the moraine that follows the same course, and runs at right angles over the older moraines of the Lake Michigan lobe. The tract in Tippecanoe and Montgomery counties, that in south Marion county, and that in Henry and Randolph counties, in the eastern part of the state, are associated with the terminal moraines that form a broad loop with the West White river basin lying in its axis. In western Ohio the belt is intimately associated with a moraine that bordered the Miami lobe of the ice sheet, and the south-trending portion in eastern Logan and Champaign counties lies on the western margin of the Scioto lobe.

The relationship of these tracts to terminal moraines is very clear and specific. They constitute marginal phenomena of the ancient ice sheet. Their distribution completely excludes their reference to floating ice, for they not only undulate over the surface utterly negligent of any horizontal distribution, but they are disposed in loops in crossing the basins of the region, and the convexities of these loops are turned down stream. These basins for the most part open out in southerly or westerly directions which makes it improbable that ice-bearing bodies of water occupied them. But if this were not fatal, certainly the fact that the convexities of the boulder belts are turned down stream and cross the centers of the basins is precisely contrary to the distribution they must have assumed if they were due to floating ice in bodies of water occupying the basins. I hold it, therefore, to be beyond rational question that these tracts were deposited as we find them by the margins of the glacial lobes that invaded the region.

If these boulder belts were of the same nature as the average boulders of the till-sheets beneath them, then the simple fact of unusual aggregation might be plausibly referred to the accidents of gathering and deposition. But they are very clearly distinguished from the average boulders of the till by several characteristics.

1. They are superficial. Sometimes they rest completely on the surface, sometimes they are very slightly imbedded, sometimes half buried, sometimes they protrude but a slight portion, and sometimes they are entirely concealed, but lie immediately at the surface. In all cases the

aggregation is distinctly superficial. Where they are buried, the burying material is usually of different texture and composition from the subjacent till, and appears to be distinct in origin from it. The superficiality of the tract is very obvious almost everywhere, and is especially so in regions where the subjacent till is of the pebble-clay rather than boulder-clay order, for the comparative absence of boulders below emphasizes the contrast. Throughout most of the region the subjacent till is not of a very bouldery type, so that the distinction is generally a marked one.

2. The boulders of the belts are almost without exception derivatives from the crystalline terranes of Canada. Those of the great tract especially under consideration were derived from the typical Huronian rocks of the region north of Lake Huron, and from granitic and gneissoid rocks referable to the Laurentian series of the same region. These last, however, cannot be sharply distinguished from the granitic rocks derived from other parts of the Laurentian terrane. The Huronian rocks are very easily identified because of the peculiarities of some of the species. Among these the one most conspicuously characterized is a quartz-and-jasper conglomerate. The matrix is usually a whitish quartzite. This is studded with pebbles of typical red jasper and of duller rocks of jasperoid nature, which grade thence into typical quartzite pebbles. With these are mingled crystalline pebbles of other varieties. Another peculiar erratic comes from the "slate conglomerate" of Logan. It consists of a slaty matrix through which are scattered rather distantly pebbles of granitic, quartzitic and other crystalline rocks. This is one of the forms of the "basal conglomerate" of Irving. Other varieties of this "basal conglomerate" are present. In addition to these very peculiar rocks, a quartzite of a very light greenish semi-translucent hue has a wide distribution along the tract. It is readily distinguishable from the numerous other quartzites of the drift of the interior. Some years since, on returning from my first field examination of a portion of this belt, I sent a typical series of chips from the characteristic erratics to Professor Irving, who had recently returned from the study of the original Huronian region. He returned a suite of chippings that matched them perfectly throughout, all of which were taken *in situ* in the region north of Lake Huron.

Among the boulders of the belt are occasionally found specimens of impure limestone or of limy sandstone that might perhaps be referred doubtfully to some member of the paleozoic series; but on the other hand,

might with equal or greater probability perhaps be referred to the similar rocks of the Huronian series. These are quite rare, never forming, so far as my observations go, as much as one per cent. of the series. In the several definite enumerations made to determine the percentage of the doubtful specimens, the result never exceeded a fraction of one per cent. In the most extensive enumeration the result was about one-half of one per cent. Aside from these doubtful specimens there are practically no boulders in the belts that can be referred to any of the paleozoic rocks that intervene in the 500 miles between the parent series north of Lake Huron and the tract over which the boulders are now strewn. Occasionally there may be seen erratics from the paleozoic series at or near the surface, but they are not usually so disposed on the surface as to appear to be true members of the superficial boulder tract. There is, therefore, the amplest ground for the assertion that these boulder tracts are of distant derivation, and that they are essentially uncommingled with derivatives from the intermediate region.

3. The boulders of this series are much more angular than those of the typical till sheets. Some of them, indeed, are rounded, but the rounding is generally of the type which boulders derived by surface degradation and exfoliation present. They rarely have the forms that are distinctively glacial. Quite a large percentage are notably angular, and have neither suffered glacial rounding nor spherical exfoliation. Some few are glacially worn and scratched, but the percentage of these is small.

The tracts therefore present these four salient characteristics: (1) the boulders are derived from distant crystalline terranes (400 to 500 miles) and are essentially uncommingled with rock from the intervening paleozoic terranes; (2) they are essentially superficial, and the associated earthy material has a texture differing from that of the subglacial tills; (3) they are notably angular and free from glacial abrasion, except in minor degree; (4) the tracts are so associated with terminal moraines and so related to the topography of the region, that there is no rational ground for doubt that the boulders were borne to their present places by the glaciers that produced the correlative moraines.

In contrast to these superficial boulder formations, the till sheets below are made up of a very large percentage of glacial clay whose constitution shows that it was produced in part by the grinding down of the paleozoic

series. In this are imbedded boulders and pebbles that were derived from the paleozoic series as indicated by their petrological character, and, in many instances, demonstrated by contained fossils. While a small part of the boulders contained in the till are angular or but slightly worn, the larger part are blunted, bruised, scratched and polished by typical glacial action. This obvious grinding of the boulders, taken in connection with the clay product resulting from the grinding, affords a clear demonstration that the deposit was produced at the base of the ice by its pushing, dragging, rolling action.

The two formations, therefore, stand in sharp contrast; the one indicating the passive transporting action of the ice in bearing from their distant homes north of the lakes the crystalline boulders and dropping them quietly on the surface, the other indicating the active dynamic function of the ice in rubbing, bruising and scoring the material at its base. The one seems to me a clear instance of englacial and superglacial transportation; the other an equally clear example of subglacial push, drag and kneading.

Now if it were the habit of an ice-sheet of this kind to carry material from its bottom to the surface by internal movement, it would seem that the distance of 400 to 500 miles which intervened between the source of the crystallines and the place of their deposit would have furnished ample opportunity for its exercise, and that there would have been commingled with the englacial and superglacial material many derivatives from the intermediate region, and these derivatives should have borne the characteristic markings received by them while at the base of the ice. The very conspicuous absence of such commingling, and the absence or phenomenal rarity of anything that even looks like such a commingling, appears to me to testify in quite unmistakable terms to the distinctness of the methods of transportation. In view of the great territory over which this particular belt is spread, and the greater territory which is embraced in the other tracts not here specially considered, there is left little ground for doubt that this distinctness of englacial from basal transportation was a prevailing fact and not an exceptional one. This is supported by concurrent evidence derived from the territory west of Lake Michigan. This territory unfortunately does not bear erratics that have equally distinct characteristics, but, so far as my observation goes, the phenomena are alike throughout. I am therefore brought to the conclusion that, in the interior at least, there was no habitual lifting of boulders from the base of the ice

sheets to the surface, nor any habitual commingling of basal with englacial and superglacial material, except, of course, as it took place by virtue of the falling of the latter through crevasses to the base, and by mechanical intermixture of the two at the edge of the ice.

The amount of englacial till under this view is little more than that which was lodged in the body of the ice in its passage over the knobs and ridges of the hilly and semi-mountainous regions of the north. To this is perhaps to be added occasional derivatives from the more abrupt prominences of the paleozoic region and the superficial dust blown upon the ice from the surrounding land, which was probably the chief source of the silty material intermingled with the superficial boulders. The total amount is thus quite small, though important in its significance.

The eskers and kames of the region are made up of derivatives from the basal material as shown by (1) the local origin of the material in large part, (2) the mechanical origin of the sands and silts, (3) the not infrequent glacial markings of the pebbles and boulders, and (4) the disturbed stratification of the beds.^[12] If I am correct in respect to the kind and amount of the englacial and superglacial material, it is obvious that eskers and kames, such as are found in the interior, could not be derived from englacial or superglacial sources. The term englacial as here used does not include such materials as may be lodged in the basal stratum of the ice and brought down to the actual bottom by basal melting.

^[12] See "Hillocks of Angular Gravel and Disturbed Stratification," Am. Jour. Sci. Vol. XXVII., May 1884, pp. 378-390.

The conclusions drawn from the phenomena of the plains of the interior are not necessarily applicable to more hilly or mountainous regions.

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STUDIES FOR STUDENTS.

DISTINCT GLACIAL EPOCHS, AND THE CRITERIA FOR THEIR RECOGNITION. [\[13\]](#)

[\[13\]](#) Read before the American Geological Society at Ottawa, Dec., 1892.

I. INTRODUCTION.

It has long been evident that writers on glacial geology are not at one concerning some of the important questions which underlie the interpretation of the history of the glacial period. Certain recent publications have served to emphasize the differences between them. There are two questions, at least, concerning which there must be agreement, or at any rate a common understanding, before existing differences can be eliminated or justly evaluated. When the answers to these questions have been agreed upon, or when the positions of the contending parties are clearly understood, it may be found that some of the apparent antagonisms have no better basis than differences in definition. Stated interrogatively, the two questions referred to are these: 1. What constitutes a glacial epoch as distinct from other glacial epochs? and 2. What are the criteria for the recognition of distinct glacial epochs, if such there were?

II. THE IDEA OF A GLACIAL EPOCH.

It is conceivable that, after the development and extension of a continental ice-sheet, it might be wholly wasted away. The maximum extension of such an ice-sheet would mark the culmination of a glacial epoch. If subsequently another ice-sheet of considerable dimensions were accumulated, its development and extension would constitute a second glacial epoch. These successive ice-sheets might be so related to each other in time, in position, and in the sequence of geological events, as to be

regarded as separate epochs of the same glacial period.^[14] On the other hand they might be so widely separated from each other in time, in position, and in the sequence of geological events, as to make their reference to separate glacial periods more appropriate. In any case their separation would be sufficiently marked to necessitate their reference to separate ice epochs. So far we believe there would be no disagreement.

^[14] The terms period and epoch are here used in the sense in which they have been used most commonly in the literature of glacial geology in the United States.

If, instead of entirely disappearing, the first ice-sheet suffered great reduction of volume and area, and if this reduction were followed by a second great expansion of the ice, might the time of such expansion be regarded as a second glacial epoch of the common glacial period? To this question, too, as thus stated, we apprehend there would be but one answer, and that affirmative.

It seems certain that the edge of the continental ice-sheet was subject to more or less extensive oscillations, as are the ends of glaciers and the edges of ice-sheets to-day. How much of an oscillation is necessary, and under what attendant conditions must it take place, in order that the recession of the ice-edge shall mark an interglacial and its re-advance a distinct glacial epoch? When the question takes this specific form, and when inquiry is made concerning the quantitative value of the different elements entering into the problem, we reach the battled ground. It is the battled ground, partly because it is the ground of misunderstanding. It is the ground of misunderstanding, partly because glacialists are not agreed as to the meaning of certain terms in common use by them.

Four elements seem to enter into the idea of an ice epoch as distinct from other ice epochs. These are (1) the distance to which the ice retreated between successive advances; (2) the duration of the retreat, or the time which elapsed between successive ice extensions; (3) the temperature of the region freed from ice during the time between maxima of advance; and (4) the intervention between successive advances, of changes interrupting the continuity of geological processes.

(1.) It would be arbitrary to name any definite distance to which the ice must recede in order to constitute its re-advance a distinct ice epoch. It would be not so much a question of miles as a question of proportions. Considering this point alone, we presume it would be agreed that an ice-sheet should have suffered the loss of a very considerable proportion of its mass, and that it should have dwindled to proportions very much less than those subsequently attained, before its re-advance could properly be called a separate glacial epoch. To be specific, if the North American ice-sheet, after its maximum extension, retreated so far as to free the whole of the United States from ice, we should be inclined to regard a re-advance as marking a distinct ice epoch of the same glacial period, if in such re-advance the ice reached an extension comparable with that of the earlier ice-sheet. Especially should we be inclined to refer the second ice advance to a second glacial epoch, if it, as well as the preceding retreat, were accompanied by favoring phases of some or all the other three elements entering into the notion of a glacial epoch. In this statement we do not overlook the fact that a northerly region—as Labrador or Greenland—might be continuously covered with ice throughout the time of the two glaciations of the more southerly regions. But this is not regarded as a sufficient reason for discarding the notion of duality. Greenland has very likely been experiencing continuous glaciation since a time antedating that of our first glacial deposits. The renewal to-day of glaciation comparable in extent to that of the glacial period would certainly be regarded as a distinct glacial epoch, if not a distinct glacial period, even though Greenland's glaciation may not have been interrupted. Scandinavia and Switzerland have probably not been freed from ice since the glacial period. Their snow and ice fields are probably the direct descendants of the ice fields of the glacial period. An expansion of the existing bodies of ice in these countries to their former dimensions, would constitute a new glacial epoch, if not a new glacial period. Analogous subdivisions in pre-Pleistocene formations have been frequently recognized.

(2) The application of the time element is hardly susceptible of quantitative statement. We are inclined to think that it would be generally agreed that, with a given amount of recession of the ice, its re-advance would be more properly regarded as a distinct glacial epoch if the interval which had elapsed since the first advance were long. Whether a longer time

between the separate advances might reduce the amount of recession necessary in order to constitute the second advance a second epoch, we are not prepared to assert; but we are inclined to think it might.

(3) The third element is perhaps somewhat more tangible than the second. If, during the retreat of the ice, the climate of a region which was twice glaciated became as temperate as that of the present day in the same locality, we should be inclined to regard the preceding and succeeding glaciations as distinct ice epochs, especially if the intervening recession were great and its duration long.

Unfortunately for simplicity and ease of determination, there are difficulties in determining with precision how far the ice retreated between successive maxima of advance, how long the interval during which it remained in retreat, and the extent to which the climate was ameliorated, as compared with that which went before and that which followed.

(4) If changes of any sort which interrupt the continuity of geological processes intervened between successive maxima of advance of the ice, the separation of the later advance from the earlier, as a distinct ice epoch, would be favored. How great the intervening changes should be in order to constitute the re-advance a distinct ice epoch, is a point concerning which there might be difference of opinion. But it is altogether possible that such changes might intervene as alone to give sufficient basis for the separation. Orographic movements, resulting either in continental changes of altitude or attitude are among the events which might come in to separate one ice epoch from another. Changes of this sort have often furnished the basis for the major and minor divisions of time in other parts of geological history, so that there can be no question as to their adequacy, if they were of sufficient magnitude. We hold that the intervention of orographic or other important geologic changes might reduce to a minimum the amount of recession, the duration of the recession, and the warmth of the intervening climate necessary to constitute the separate ice advances separate ice epochs. The absence of great orographic or other changes in glaciated regions between successive advances of the ice would be no proof that such advances should not be regarded as separate epochs. Divisions of equal importance have often been made without evidence of such changes.

From the foregoing discussion, brief as it is, it will be seen that within certain narrow limits the definition of a glacial epoch, as distinct from other glacial epochs, must be more or less arbitrary. It is less important that an arbitrary definition should be accepted, than that the same meaning should be attached to technical terms in common use among geologists. In the interest of harmony and of a common understanding, and without the violation of any truth of science, we believe it would be well if the conception of a glacial epoch, as framed by those who are our leaders in position and in fact, were made the basis for our usage of the term.

III. THE CRITERIA OF DISTINCT GLACIAL EPOCHS.

If there have been differences of opinion concerning the nature of ice epochs, as distinct from each other and from ice periods, there has been a failure to adequately apprehend the nature, the extent, and the meaning of the real criteria on which the final recognition of separate ice epochs, if such there were, must be based.

Such criteria are several in number. They are of unequal value. In some instances a single one of them might be quite sufficient to establish the fact of two ice epochs. In other cases, single criteria which might not be in themselves demonstrative, have great corroborative weight, when found in association with others. In all cases, much discretion must be used in the interpretation of these criteria. They may be enumerated under several specific heads.

(1) *Forest Beds.* Beds of vegetal deposits or old soils are frequently found between layers of glacial drift. This is one of the criteria most commonly cited, because it is of common occurrence and easy of recognition. The advocates of the unity of the glacial period maintain that such beds of organic matter might become interbedded with morainic debris during minor oscillations of the ice's edge. The phenomena of existing glaciers make it evident that forest beds or soils might be enclosed by the deposits of an oscillating ice edge. By repeated oscillations of the ice's edge during the general retreat of the ice, such vegetal beds might become interstratified with glacial drift more or less frequently over all the area once covered by the ice, and from which it has now disappeared. The mere

presence of vegetable material between beds of drift is therefore no proof of distinct ice epochs. This does not destroy the value of the vegetal beds as a criterion for the recognition of distinct ice epochs, but it makes caution necessary in its application. It does not follow that, since *some* inter-drift forest-beds do not prove interglacial epochs, *none* do. The question is not how forest-beds might originate, but how existing forest-beds did originate.

Where the plant-remains found in the relations indicated are so well preserved as to make identification of the species possible, we have a means of determining, with some degree of accuracy, the climatic conditions which must have obtained at the place where the plants grew during the time of their life. If these interbedded plant-remains are of such a character as to indicate a temperate climate, we can not suppose that they grew at the immediate edge of the ice, and therefore that they were buried beneath its oscillating margin. To be specific, if the inter-drift plant remains in any given locality of the area once covered by ice are such as to indicate a climate *as warm as the present in the same locality*, the ice must have receded so far to the northward that its re-advance might, in our judgment, appropriately be regarded as a separate ice epoch.

It has been suggested in opposition that temperate conditions may obtain even up to the edge of the ice, and that interbedded vegetal remains indicating temperate climate do not prove any considerable recession of the ice. The phenomena about existing glaciers have been appealed to in support of this demurrer. But the objection is not well taken. The climatic conditions which obtain about the borders of small, local glaciers, are not a safe guide as to climatic conditions which obtained about the margin of a continental ice-sheet, any more than the climatic conditions which obtain about a small inland lake are a safe criterion as to the climatic conditions about a sea-coast. The general principles of climatology, as well as specific facts concerning plant distribution, seem to us to indicate that the climate about the border of a continental ice-sheet must have been arctic.

It is evident that the greater the distance north of the overlying drift remains of temperate plants are found, the more conclusive becomes the evidence. Plant remains indicating temperate climate at the very margin of the drift sheet which overlies them, would be less conclusive than similar evidences one hundred miles to the northward. It might be difficult to prove

in any given instance that the ice which deposited the drift overlying plant remains advanced one hundred miles, or any other specific distance, south of any particular underlying forest bed. If the forest bed were continuous for the whole distance, the case would be clear. It would also be conclusive if the continuity of the drift overlying a forest bed at any point with that of a remote point to the south, could be demonstrated. In spite of these difficulties in its application, the vegetal beds constitute a valuable criterion in making the discriminations under consideration, when they are properly applied. Under proper circumstances the criterion may be conclusive when taken alone, and it may have corroborative significance when not itself conclusive.

The absence of forest beds and of all traces of vegetal deposits whatsoever between beds of drift, is no proof of the absence of recurrent ice epochs, since the second advance of the ice might have destroyed all trace of the preëxistent soil and its vegetal life. It is always possible, too, that such beds exist, even if they have not been discovered. It would have been anticipated that they would not be abundant, or wide spread. The absence of forest beds is therefore at best no more than negative evidence.

(2) *Remains of Land Animals.* Bones of mammalia or remains of other land animals, occurring in relations similar to those in which forest beds occur, may have a like significance. Their value as a criterion of separate glacial epochs is subject to essentially the same limitations as forest beds.

(3) *Inorganic Products formed during a time of Ice Recession.* The recession of the ice after a maximum of advance would leave a land surface more or less affected with marshes and ponds. In such situations, bog iron ore might accumulate, if conditions were favorable. Such ore beds, buried by the drift of a later ice advance, would have a significance comparable to that of forest beds, except that they would give less definite information as to climate, and would be correspondingly less trustworthy. Should such ore beds be found in such relations as to prove that the underlying and overlying bodies of drift were deposited by ice sheets which extended great distances further south, their significance would be enhanced. From the thickness of the ore beds some inference might be drawn as to the length of time concerned in their accumulation. But because of the variable rate at which bog ore may accumulate, such inference should be used with caution.

Concretions of iron oxide might be formed in the marshes or in ill-drained drift areas where accumulations of greater extent were not made. A subsequent incursion of the ice might incorporate these nodules with its drift, wearing and striating them as other stones, and depositing them as constituent parts of the later drift. Such iron nodules in the later drift would mean a recession and re-advance of the ice with some considerable interval between, although not necessarily an interval sufficiently warm or long to be regarded as an interglacial epoch.^[15] Calcareous concretions, like those of the loess, would possess a like significance, in like relations. While in themselves these inorganic products of a time of ice recession might fail to be conclusive of separate ice epochs, they might have much corroborative significance when associated with other phenomena. An inter-till iron ore bed, associated with a forest bed which indicated a warm climate, would be most significant.

[15] This point concerning iron nodules was suggested to the writer by Mr. W. J. McGee.

The absence of knowledge of ore beds between sheets of till, and the absence from an upper bed of till of concretions of iron and lime carbonate formed during a recession of the ice, would be no proof that interglacial epochs did not occur. These products were probably formed in relatively few localities. They stood good chance of destruction at the hands of the returning ice, and they may exist, where they have not been discovered, or where their significance has not been understood. Their absence is at best no more than negative evidence.

(4) *Beds of Marine and Lacustrine Origin.* If between beds of glacial drift there be found beds of lacustrine or of marine origin, such beds would indicate a recession of the ice during their time of deposition. Their position would be a minimum measure of ice recession. If such lacustrine beds contain organic remains, they will bear testimony concerning the climatic conditions which existed where they occur, at the time of their deposition. If the fossils in such beds denote a temperate climate, or a climate as mild as that of the present day in the same region, the ice must have receded so far to the northward as, in our judgment, to constitute its re-advance a distinct ice epoch. This line of argument may be even stronger than that drawn from remains of terrestrial life, since the ice would probably affect the temperature of the sea to greater distances than that of the land, and affect it to a greater degree within a given distance. The argument becomes stronger the further north the inter-drift marine and lacustrine deposits occur, since the ice must always have receded to a position still further north. If marine or lacustrine beds lying far north of the later ice limit contain proof of temperate climate, the argument becomes conclusive.

The absence of marine and lacustrine deposits between beds of drift, would be no proof that interglacial epochs did not occur. Lacustrine beds could be made only where there were lakes, and lakes would be the exception rather than the rule. Marine beds in similar positions would rarely be known, except where a definite succession of changes of level has taken place. Both classes of deposits, if once formed, would be subject to destruction by the over-riding ice of a later epoch, if such there were. Neither would be likely to be preserved at all points where formed, and

both may exist at many points where their existence is not known. The absence of these beds is at best no more than negative evidence.

(5) *Beds of Subaërial Gravel, Sand and Silt.* Layers of stratified drift between layers of ground moraine are of common occurrence in many regions. Under ordinary conditions their existence is not regarded as evidence that the underlying and overlying tills are to be referred to separate ice epochs. But it is conceivable that beds of stratified drift may, under the proper circumstances and relations, be strong evidence of separate ice epochs. The last stages of ice work in the glacial period were accompanied, in many regions, by the deposition upon adjacent land surfaces, of extensive bodies of gravel and sand, washed on beyond the ice by waters issuing from it. Except in valleys through which strong currents coursed, such deposits were apparently not carried far beyond the edge of the ice. But as the edge of the ice withdrew to the northward, sand plains may have extended themselves in the same direction, by additions to their ice-ward faces. It is conceivable that the process of subaërial plain building at the edge of a receding phase of ice, might be carried so far under favorable circumstances, as to result in the construction of plains of great extent. In this event, a subsequent ice-advance might overspread such plains in such wise as to bury, without destroying them, though such a course of events would certainly be exceptional. In order to constitute the interstratified gravel and sand evidence of separate ice-epochs, its continuity for great distances between beds of till, and in the direction of ice movement, would need to be demonstrated. In themselves, these beds, under the conditions indicated, would simply be a minimum measure of the amount of ice recession between the deposition of the underlying and overlying bodies of till. It is hardly likely, though possible, that the continuity of interbedded gravel and sand could be proved for a sufficient distance north of the southern limit of the less extensive bed of ground moraine, to alone constitute evidence of a recession of ice great enough to make it necessary to refer its re-advance to a new epoch. Beds of silt in like relations, deposited by waters beyond the edge of the ice, would have a like significance so far as the question here under consideration is concerned. Such beds of stratified drift might sometimes have corroborative value when their testimony, taken by itself, is inconclusive. If, for example, their

surfaces are marked by forest beds, and especially by forest beds whose plants denote a warm climate, the association becomes most significant.

In view of what has been said, it is evident that the absence of beds of subaërially stratified silt, sand, and gravel, between beds of till can not be brought in evidence against separate ice epochs. It would rarely be true that topographic and hydro-*graphic conditions would make possible the construction of plains of sufficient extent to serve as criteria for the purpose here indicated, and few of those formed would escape such a degree of destruction as to leave them demonstrably continuous. There is also the further possibility that such beds exist, even though their continuity be not known. To prove the continuity of a buried bed of stratified and incoherent drift, even if it existed, would be a most difficult task.

(6) *Differential Weathering.* If, after covering a given region, the ice retreated, the drift which it left in the area which it previously covered would be subject to oxidation, leaching and disintegration. The depth to which this oxidation, leaching and disintegration would extend, would be dependent upon the length of time during which the drift was exposed, and upon the climate which affected the region during its exposure. The longer the exposure and the warmer the climate, the deeper would the weathering extend. If, subsequently, the ice extended over the same region, it might, in some places, override and bury the old surface without destroying it. The earlier oxidized and leached drift would thus come to be buried by the newer, unoxidized, unleached drift. If, therefore, beneath the newer drift of any given locality there be found a lower drift, the surface of which is oxidized and leached to a considerable depth, the evidence is strong that the lower drift was exposed for a long period of time before the upper drift was deposited upon it. Within certain limits a similar result might be brought about, it is true, if the ice, after having reached a certain maximum stage of advance, were to retreat for a short distance only and there remain for a very long period of time. A subsequent minor advance might bury the oxidized surface of the drift beyond the position of the long ice-halt. Under these conditions, the climate which would have obtained in the area of the drift exposed during the minor retreat would have been cold, and oxidation, leaching, and disintegration would have proceeded slowly. If they reached considerable depths, the time involved must have been very long. If this surface of oxidized and leached and disintegrated drift were found to reach

far to the northward beneath the layer of newer and upper drift, it would indicate a great recession of the ice. We maintain that if it were found sufficiently far north of the margin of the overlying drift, and if its depth were sufficiently great, extending well down below any possible accumulation of superglacial till, it might be a positive criterion of so great a recession of the ice, protracted through so great an interval of time, as to constitute its new advance a separate ice epoch.

There is much reason to believe that the soil developed under the influence of a warm climate differs in some respects from one developed from similar material under other conditions. The well-known fact that red and reddish soils are especially characteristic of low latitudes and warm climates is significant. If therefore a soil developed on the surface of one sheet of drift and buried by another, be found to possess, in addition to unmistakable marks of long exposure, the peculiar marks which seem to be characteristic of soils developed under high temperatures, the argument gains in strength.

This argument from oxidation and weathering has another application. If in a later advance, following a protracted recession, the ice-sheet failed to reach the limit of its earlier advance, there would remain an area of drift deposited by the first ice-sheet, outside the drift deposited by the later. Now if the time interval between these two advances was great, and especially if during this interval the climate was mild, the oxidation and weathering of the older drift surface would be markedly different in degree from that of the newer. If, under these circumstances, the surface of the older sheet were found to be weathered and oxidized and reddened up to the border of the newer drift sheet, and if here there were found to be a sudden change in the character of the surface of the drift so far as depth and degree of oxidization and weathering is concerned, we should have strong evidence that the one sheet of drift was much older than the other. The statement sometimes urged that the drift which was deposited near the edge of the greatest ice advance would be largely made up of the residual materials which occupied the surface invaded by the ice, would not meet the case. For if it be granted that this statement is qualitatively good, we should find the greatest degree of weathering and oxidation at the extreme margin of the drift, and it should be found to be less and less on receding from this margin. There would in this case be no sudden transition from a deeply weathered and oxidized

surface, to one which is fresh and unoxidized, along a definite line. We maintain that if the whole of the drift deposits are referable to one epoch, there should be no sudden transition in the surface of the drift from that which is deeply weathered to that which is not, the one surface being separated from the other by a definite and readily traceable line.

It has been urged against the criterion of differential weathering that superglacial material is or may be thoroughly oxidized before its deposition, and that a layer of oxidized drift between layers of till may be no more than superglacial debris deposited during a minor recession of the ice.^[16] We believe that this attempt to eliminate the value of this criterion rests partly on an exaggerated idea concerning the amount of superglacial material, but more especially on a failure to apprehend the real meaning of the argument for the validity of the criterion, and upon a failure to note the limitations imposed upon it by its advocates. It is not affirmed that a layer of oxidized drift between beds of unoxidized drift is *per se* proof of two glacial epochs; but it is affirmed that if such layer of weathered drift can be shown to extend far below any possible superglacial till, into the subglacial till below, in such wise as to indicate that it is the result of subaërial exposure in a warm climate subsequent to its deposition and prior to the deposition of the overlying till, it constitutes the best possible evidence of an interglacial epoch, especially when accompanied by the corroborative testimony of other criteria. It is further affirmed that if the second sheet of drift failed to reach the limit of the first, and if the drift which was deposited by the first and never covered by the second ice-sheet, is more thoroughly and more deeply weathered than that deposited by the second, and especially if the two types of drift surface meet along a definite and readily traceable line, the argument becomes, in our judgment, irrefragable. In its application, this criterion would be infallible only in the hands of one who could distinguish between superglacial and superglacially oxidized material on the one hand, and material subaërially weathered after its deposition, on the other.

^[16] This point was urged at the reading of the paper at Ottawa, by Prof. C. H. Hitchcock, Mr. Upham, and others.

In circumstances and relations where the weathering of the drift is not in itself conclusive, it might still have corroborative value in association with other lines of evidence.

The absence of an oxidized and disintegrated zone of drift below a superficial layer which is not oxidized, would be no proof that there were not distinct ice epochs, since the ice of any later epoch, if such there were, might have planed off the surface of the drift left by its predecessor to the depth of the weathering. The preservation of such surfaces after a second ice invasion must be regarded as the exception rather than as the rule. There is always the possibility, too, that an oxidized and weathered zone marking the surface of an older drift sheet exists, where excavations have not opened full sections of drift to view. The absence of weathered zones of drift beneath the surface, or the absence of knowledge of their existence, is therefore at best no more than negative evidence. The absence of greater weathering of the drift outside the limit of the drift supposed to belong to a later epoch, would be positive evidence against the reference of the two sheets of drift concerned to different epochs.

A specific part of the above line of evidence may be separately mentioned. One phase of weathering is the disintegration of boulders, and this is a point which can be readily applied even by those who are not geologists. If the boulders of one region are much more commonly disintegrated than those of another, and if the two regions are separated from each other by a well-marked boundary line, the inference lies close at hand that the boulders in the one case have been much longer exposed to disintegrating agencies than in the other. It is no answer to this argument to say that the materials lying at the very front of the drift deposits contain boulders which were derived from the disintegrated rock over which the ice has passed, and that they were therefore in a less firm state at the outset. In many cases these boulders have come from great distances, and coming from great distances they must have come in a firm and solid state, else they could not have suffered such extensive transportation, except indeed their position was superglacial throughout their whole journey. This argument has equal force when applied to the area covered by the two sheets of drift where two exist. If within the region of drift under investigation we find a surface layer of greater or less depth, the boulders of which are hard and fresh, and if beneath this we find another layer of drift, the stony material of which is largely disintegrated, at least in its upper parts, we have good evidence that the surface bearing the disintegrated boulders was exposed for a considerable length of time before the deposition of the overlying drift,

which carries fresh boulders. Since the disintegration of boulders is only one phase of weathering, the limitations of this argument are identical with those already noted in connection with the general argument from differential weathering.

(7) *Differential Subaërial Erosion.* If the drift deposited by one ice-sheet were to be exposed for a considerable interval of time, and if the ice in its subsequent advance failed to reach the limit of its first invasion, the two areas should show different amounts of subaërial erosion, since the one has been exposed to the action of air and water much longer than the other. The line which marks the limit of the later ice invasion should be the line of more or less sudden transition from an area without, where stream erosion has been greater, to an area within, where stream erosion has been less.

The point here made can not be met by the suggestion that the greater erosion of the outer area was effected by the water issuing from the ice which had retreated to the position now marked by the border of the area of the lesser erosion. So far as we know, such waters would be depositing, not eroding. Furthermore, much of the erosion of the outer area would have such relation to drainage lines that waters issuing from the ice could never have reached the localities where it is shown.

If the outer and older drift be found to have suffered ten times as much stream erosion as the inner and newer, it is fair to assume that it has been exposed something like ten times as long, if the conditions for erosion are equally favorable in the two regions. The argument has especial weight if it can be found that beneath the newer drift the surface of the older is such as to indicate that it was deeply eroded before the newer was placed upon it. The argument is stronger the farther from the margin of the newer drift such erosion on the surface of the underlying older drift can be proved to have taken place. In other words, if, in addition to the greater surface erosion of the older drift sheet as now exposed outside the limit of the newer drift, we find a notable unconformity between the newer and the older drift, and especially if this unconformity lie far enough north of the margin of the newer drift, the argument becomes conclusive.

When differential erosion and drift unconformities are not in themselves conclusive, they may have great corroborative value in conjunction with

differential weathering, forest beds, or other indications of separate ice epochs.

The absence of observable unconformity between sheets of drift would be no proof that there were not distinct and widely separated ice epochs, since the later ice invasion might have so far modified the surface which it transgressed, as to destroy all patent evidences of unconformity. It would have been anticipated that distinct unconformities in the drift would be rare, even if there were distinct ice epochs, for the same reason that weathered zones and forest beds would be rare. But if the drift which lies outside a line supposed to mark the limit of a sheet of drift belonging to a later ice epoch, be not more eroded than that which lies within such line, the absence of greater erosion in the outer drift is positive evidence against the reference of the drift of the two areas to distinct ice epochs, if conditions for erosion in the two areas are equally favorable.

(8) *Valleys Excavated Between Successive Depositions of Drift.* A closely related, but not identical, point may be found in the extent of the valley excavations which can be proved to have taken place between the deposition of the earlier and later drift. We do not refer to valleys excavated in the drift especially, but to those excavated in other formations as well. If it can be shown, for example, that after the deposition of an earlier drift sheet, and before the deposition of a later, valleys were excavated which extended not merely into the drift itself, but far beneath the drift into the underlying rock, these valleys would be conclusive evidence of a long interval between the deposition of the two bodies of drift. The argument is of especial force when such excavations in the rock beneath the drift can be shown to have taken place at great distances within the margin of the newer drift. For valleys in such situations imply that the ice had receded at least as far to the north as they lie, during the interval between the two drift depositions, and may be so situated as to show that the ice had wholly left the drainage basin where they occur.

The absence of evidences of deep valley excavations in any given region during a supposed interglacial epoch, is no proof that such interval did not exist. The conditions may not have been everywhere favorable for erosion within the limits of any narrowly circumscribed area, and the absence of interglacial valleys would be only negative evidence against an

interglacial epoch. The absence of such evidence everywhere would bear against the existence of an interglacial epoch of much duration in such wise as to be more than negative evidence.

(9) *Different Directions of Movement.* If, after its maximum advance, the ice suffered merely a minor recession and then remained stationary, or nearly so, for a time, the general direction of its movement in a subsequent advance would probably be essentially the same as in the earlier. But if, after its maximum advance, the ice receded to a great distance, and especially if it entirely disappeared, a subsequent ice-sheet might have a very different direction of movement, since its center of accumulation and dispersion might be very different. It is conceivable that this center might shift during the history of a single ice-sheet. In this case there should be a gradual change in the direction of ice movement, not an abrupt one. If, therefore, there be found one sheet of drift made by an ice movement in one direction, overlaid by another sheet of drift deposited by ice moving in a very different direction, with an abrupt transition between them, such drift sheets would be presumptive evidence of distinct ice epochs. An exception would need to be made in the case of drift sheets along the margins of confluent or proximate ice lobes. In such cases, if the one lobe temporarily secured the advantage of the other, drift beds formed by movements from opposite directions might be found in vertical succession, without being evidence of separate ice epochs.

It is no part of the purpose of this essay to point out the difficulties which might arise in the application of this criterion of diverse directions of ice movements. It is possible that gradual changes in the direction of movement might leave records which would seem to indicate abrupt changes instead. This possibility makes care necessary in the application of the criterion, but does not destroy its value. When not itself conclusive, this criterion may be so associated with differential weathering, differential erosion, forest beds, etc., that their combined testimony makes but one conclusion possible.

The absence of evidence of radically diverse directions of movement during the time of deposition of the various sheets of drift, would be no proof that there were not distinct epochs. In the first place, the movements of different epochs might be harmonious—a condition of things more

probable than any other if the more common views of the causes of glaciation be correct. In the second place, if the movements were diverse, the deposits might still be so similar that their differentiation, when the one is buried, might not be easily made. In the third place, the later ice might have so far incorporated the older drift material with that which belonged more properly to it, as to have destroyed all definition between them.

(10) *The Superposition of Beds of Till of Different Physical Constitution.* After the retreat of an ice-sheet, the surface of the country thus discovered would be largely mantled with drift. This drift would serve to protect the underlying rock from disintegration. But where there was little or no drift, the rock surface would be subject to all the disrupting agencies which affect surface rocks. The same would be true of all rock surfaces bared by subaërial erosion after the disappearance of the ice. Under these conditions, if a second sheet of ice invaded the region in question after it had been long exposed, it would find a surface prepared to yield large boulders. The result would be the deposition of a new sheet of drift containing boulders much larger than those which would have been proper to an ice-sheet overspreading a surface but recently abandoned. If, therefore, in the upper of two layers of subglacial till, boulders of great size predominate, as compared with those of a lower homologous layer, they may be indicative of a great interval of time between the deposition of the upper and lower beds of drift. If the home of these boulders be far north of the limit of the lesser sheet of drift, the distance, as well as the duration, of the ice retreat must have been great, and the reference of the two beds of till to distinct ice epochs would be favored. The case might be so strong as to make no other interpretation possible. Where in itself inconclusive, this criterion would have corroborative significance. In its application, the discrimination of subglacial and superglacial till would be imperative.

The absence of physical dissimilarity between superposed layers of subglacial till would not be proof of the absence of separate glacial epochs. The phenomena constituting the criterion could hardly be expected to be of common occurrence. They would never be obtrusive, and may easily have escaped attention where they exist. ^[17]

^[17] The 10th criterion, in the order here named, was suggested by Mr. McGee in the discussion which followed the reading of the paper at Ottawa.

(11) *Varying Altitudes and Attitudes of the Land.* Another line of argument has to do with the altitude and attitude of the land during the deposition of various members of the drift complex. If during the deposition of one part of the drift that part of the continent covered by the outer part of the ice was low, the drainage from it would be sluggish. If the deposits of this drainage persist to the present time, we may find in their character evidence of the nature of the drainage, and therefore of the attitude of the land. If at a later time of drift deposition the glacial drainage in the same region was more vigorous, the deposits made by the glacial streams would be correspondingly coarser. In these deposits, if they persist to the present day, we should find conclusive evidence of the swiftness of the streams. If it can be shown that during the deposition of one sheet of drift drainage was sluggish, and that during the deposition of a later body of drift the drainage was vigorous, these facts are evidence of an interval between the two times of drift deposition, sufficiently long to accomplish the corresponding changes in elevation or attitude. Since such changes of altitude and attitude are generally believed to have been accomplished slowly, the interval must be believed to have been of considerable duration.

It is true that continental altitudes and attitudes might change during a single epoch of glaciation. If the change thus brought about resulted in increased slope, the more sluggish drainage of the earlier part of the epoch would be gradually transformed into the more vigorous drainage of the later part. In this case, if the evidence of both the earlier sluggish drainage and of the later vigorous drainage remain, there should also remain the evidence of the intermediate stages. If the deposits representing the intermediate condition of drainage do not exist, while those representing both extremes do, there would be the best of reason for believing that the intermediate phases of drainage did not exist during a glacial epoch, but during an interglacial epoch, when streams were not handling glacial debris, and when they were eroding rather than depositing. The deposits of the slow and of the swift drainage might occur in such relations as to prove, beyond peradventure, that intermediate stages of *glacial* drainage never existed.

If the sluggish drainage accompanied the maximum ice invasion, while the vigorous accompanied a lesser, the evidence of the swift streams might be found far north of the southern limit of the earlier drift. The farther north of the outer border of the older drift the gravel representing the vigorous

drainage of the later and minor ice-sheet occurs, the further the ice must have retreated before the change from the one type of drainage to the other was effected. On the other hand, the farther north of the limit of the later ice advance the sluggish drainage accompanying the earlier ice-sheet may be traced, the farther must the ice have receded before the changes resulting in vigorous drainage occurred. Under certain relations, the retreat of the ice might be shown to have been great enough, before the orographic movements which altered the nature of the drainage, to constitute in our judgment, a re-advance a distinct ice epoch. If for example throughout the course of a long river whose basin was largely covered with ice, there be evidence that sluggish drainage obtained during the maximum ice advance, and during all stages of the ice retreat until the basin was free from ice, and if there be evidence of a vigorous glacial drainage in the same valley at a later time, with no gradations between the two types, we have proof positive of at least a great recession, and of a considerable elevation of the land after the ice had receded beyond the limits of the drainage basin and before it again reached it in its re-advance. We hold that these phases of glacial drainage deposits may be so related to each other, to the valleys in which they occur, and to more or less distinct bodies of glacier drift, as to prove so great a recession of ice between the diverse phases of drainage deposition, as to constitute the second advance a distinct ice epoch.

The absence of evidence that the land stood at different elevations during different parts of the period of drift deposition, does not in any way militate against the theory of recurrent and distinct ice epochs. A constant attitude of the land is the thing to be assumed, until positive evidence to the contrary is adduced.

(12) *Vigor and Sluggishness of Ice Action.* If it can be shown that during one epoch of glaciation, we will say the epoch of maximum ice extension, the ice action was relatively sluggish, while during a later and minor advance its action was vigorous, the difference of action might be regarded as presumptive evidence of distinct ice epochs. Evidence of the two phases of ice action here referred to are difficult of definition, but they have been independently noted by more than one glacialist. It is true that a forward oscillation of the ice edge might be more forceful than an earlier forward movement which might have reached a greater extension. In itself,

therefore, this line of evidence can not be regarded as possessing great value.

It has been indicated that under certain circumstances, and in certain relations, some of the foregoing criteria, taken singly, may be conclusive of glaciations so distinct from each other, as to make their reference to separate epochs proper. But where the facts and relations which constitute one of the criteria are found, the facts and relations constituting one or more of the others are likely to be found as well. Where two of the foregoing criteria are found to be coexistent, their joint force is greater than that of either one. If neither one be absolutely conclusive, the two may still be, since the one may exactly meet the deficiency of the other. If three or more concurrent lines of evidence exist in any locality, the case is still further strengthened. We maintain that several of the foregoing criteria may be so related to each other and to the formations concerned, as not only to make the recognition of separate ice epochs proper, but to make the failure of such recognition altogether unscientific. Even when a single line of evidence, or when double, or triple, or quadruple lines of evidence are not absolutely conclusive in ruling out every conceivable technical escape from the conclusion that there were separate ice epochs, their cumulative and corroborative force may still be such as to carry conviction scarcely less positive than that which mathematical demonstration would afford. In the nature of the case not all of these various lines of evidence could be expected to be found in any one locality, or perhaps in any one limited geographic area, but where one occurs, some or all of the others are liable to be found under favoring circumstance. The number of criteria, and the great extent of area where they may hope for application, afford great possibilities.

From the foregoing discussion, it will be readily seen that the nature of the criteria and the limitations imposed upon their application by the difficulty of proving stratigraphic continuity in such a formation as the drift, necessitate the greatest care in their use, and reduce the value of hasty and inexpert conclusions to a minimum.

IV. AREAS WHERE THE CRITERIA FIND READIEST APPLICATION.

The foregoing criteria find their readiest application in regions where a later sheet of drift, suspected of belonging to a later ice epoch, failed to reach the border of an earlier sheet of drift, suspected of belonging to an earlier ice epoch. The 1st, 2d, 3d, 4th, 5th and 10th as enumerated above, find their application wholly within the area affected by the drift of the separate epochs, if such there were. While within this general area they may be looked for at any point, they are likely to be of rare occurrence, except along a somewhat narrow belt, say 50 to 100 miles, adjacent to the border of the lesser ice advance. The conditions for their occurrence and detection are greatly favored if the lesser drift sheet be the later. The 6th, 7th, 9th and 12th criteria might hope for application within the same belt, but especially along a narrow zone on either side of the margin of the later drift sheet. It is along this zone that the types of surface are thrown into sharpest contrast, both as to material and topography. The 8th and 11th criteria have still wider limits of application, both within and without the border of the lesser ice advance.

ROLLIN D. SALISBURY.

EDITORIALS.

It is the chief function of the national, state and provincial geological surveys to bring forth the great concrete facts relative to the structure and resources of their several fields. Within their special domains they also do an important work in the correlation of structures and formations, in the systematic aggregation of the facts, in the organizing of results, and in the development of the fundamental principles of geological science. To some extent they are permitted to do this beyond their own fields, but in the main the boundaries of these fields are the limits of their coördinations. They therefore leave a great function to be performed by some other agency in the coördination of interstate, international, and intercontinental factors. They are also restrained by their relationships to a somewhat too narrowly utilitarian public from devoting much direct attention to the solution of the deeper and broader problems that constitute the soul of science, though their contributions bear upon these in the most radical and important way. In the primary work of systematic observation, and the development of the immediate conclusions that spring therefrom, these surveys surpass all other agencies in the value of their contributions to the growth of the science, but in the secondary and ulterior work of correlation, in the synthetic aggregation and organization of results, and in the analytical and philosophical treatment of the whole, they need to be supplemented by agencies whose facilities and limitations lie in other lines, agencies whose relations and dependencies are complementary in nature. This secondary and ulterior work, in some degree, has been done by individual master students of systematic and philosophical geology, but to a very great extent it has not been done at all. It is a function which properly falls to universities, if the universities can only rise to meet it; for it is the function of universities, in the larger modern view, not only to rehearse science, nor merely even to educate young geologists, important as that is, but to develop science for science's own sake, and for its own inherent and permanent utilities as distinguished from its immediate applicabilities. To fulfill this function they must not only realize and appreciate it, but they must be equipped for field and experimental work, as well as library and

laboratory study. Ideal correlations and academic systematizing are as apt to be hindrances as helps to the progress of science. While a few of the great universities of this country and Europe have made notable advances in these directions, the universities are, on the whole, far behind the great surveys in the performance of the work which properly falls to them. This is due not so much to a lack of appreciation of the function as to the lack of facilities.

With the development of this higher function of the universities there goes a coördinate function for a university journal of geology, a journal whose special efforts shall be devoted to promoting the growth of systematic, philosophical, and fundamental geology, and to the education of professional geologists. No part of the wide domain can wisely be neglected by any journal, but there seems to be an open field for a periodical which specially invites the discussion of systematic and fundamental themes, and of international and intercontinental relations, and which in particular seeks to promote the study of geographic and continental evolution, orographic movements, volcanic coördinations and consanguinities, biological developments and migrations, climatic changes, and similar questions of wide and fundamental interest. This field is not likely to be successfully cultivated except by a systematic endeavor, pursued through a period of years, to bring together the latest and best summations of the results attained in the several national fields in a common medium, where they can be compared and discussed, and where tentative correlations will suggest themselves, out of which, in turn, working hypotheses will naturally spring, leading on to such direct investigations as the nature of each question invites. It would be presumptuous to assume that the JOURNAL OF GEOLOGY can cultivate with more than very partial success this field, but it especially invites contributions of this class.

Another phase of geology which is thought to stand in much need of active cultivation is found in the clear and sharp analysis of its processes, the exhaustive classification of its phenomena, especially on genetic bases, the development of criteria of discrimination, the more complete evolution and formulation of its principles and the development of its working methods. The recent opening of new fields of research and the rapid progress of several new and important departments of the science give peculiar emphasis to this need. The rising generation of geologists, the hope of the science, should be schooled in these latest and most critical aspects of

the science. A department of the JOURNAL, entitled "Studies for Students," has been opened for the special cultivation of this field and for its adaptation to advanced students and progressive teachers of geology. Mere elementary presentations of processes and principles are not desired, but searching and critical expositions are solicited suited to the needs of young geologists who seek the highest professional equipment, and to progressive teachers who desire the fullest practicable command of the newest developments of the subject. These contributions may not be without their value to those who have already borne a considerable part of the heat and burden of life's professional day.

It is our desire to open the pages of the JOURNAL as broadly as a due regard for merit will permit, and to free it as much as possible from local and institutional aspects. It will have the very important advantage of being published under the auspices and guarantee of the University of Chicago, and will be free from the usual financial embarrassments attending the publication of a scientific magazine. This necessarily imposes upon the local editors the immediate responsibility for its editorship. Beyond this, it is hoped that its institutional relationship will disappear entirely in an earnest effort to promote the widest interests of the science. As an earnest of this wider effort several eminent geologists, representing some of the leading universities of this country, and some of the great geological organizations of Europe, have kindly consented to act as associate editors.

T. C. C.

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Upon invitation of the World's Congress Auxiliary of the World's Columbian Exposition committees were appointed by the several sections of the American Association for the Advancement of Science at its Rochester meeting to coöperate with it in completing the organization of scientific congresses to be held at Chicago in connection with the forthcoming World's Fair. The committee appointed by the geological and geographical section consisted of Thomas C. Chamberlin, John C. Branner, Grove K. Gilbert, W. J. McGee, Rollin D. Salisbury, Eugene A. Smith,

Charles D. Walcott, J. F. Whiteaves, Geo. H. Williams, H. S. Williams and N. H. Winchell.

It has been arranged that this committee should undertake the work of preparing the scientific program for the Geological Congress. The committee have prepared a provisional schedule of topics, which they have submitted to the Advisory Council for revision. It has seemed to the committee that all contributions should be such as to have an international interest. Preferably, they should be subjects that can only be treated most advantageously in such a congress, especially those that involve the bringing together of data from different lands for comparison. The committee suggest the organization of the subjects under the following general classes:

FIRST. Such as shall show the present state of geological progress. It is believed that this can best be done by an exhibition of geological maps which shall show the latest and best results of official and other surveys. As such maps will be prepared, it is hoped, for the World's Fair, duplicates can be made at a slight expense for the use of the Congress. It is hoped that each country that has made any notable progress in mapping its geological formations will furnish for the Congress at least a general geological map, if not also special or analytical maps.

SECOND. Such subjects as bear upon continental growth and intercontinental relations. It is proposed to make this a leading line of discussion during the Congress, in the belief that there is no subject more appropriate, and that there is none which better represents the present efforts of geologists or commands a more general interest. It is hoped that analytical maps will be prepared by the geologists of the several countries representing the stages of growth of these regions in each of the great eras from the Archean to the Pleistocene, and that such analytical maps may constitute a leading feature of the several presentations. Among the subjects upon which contributions are specially invited are the following: The correlation of continental and intercontinental orographic movements and geographic accretions by sedimentation; The coördination of periods of vulcanism in the different countries; The coördination of climatic states and changes; The correlation of faunal and floral variations and migrations. It is hoped that one session may be devoted to such coördination papers bearing

upon each of the great subdivisions: viz., Archean, Paleozoic, Mesozoic, Cenozoic, and Pleistocene.

THIRD. Papers on Paleontological and Archeological Geology of international scope.

FOURTH. Contributions to Physical, Structural and Petrological Geology having international or general bearings.

FIFTH. Contributions to Economic Geology having general bearings.

SIXTH. Miscellaneous papers of especial and general interest.

The foregoing groups are intended to embrace and coördinate the list of special themes announced in the circular issued by the local committee some months since, except such as may be best suited to popular presentation, for which special provision is to be made.

It will be determined later, when the number and nature of the papers are ascertained, whether all will be arranged so as to form a continuous program, or whether sub-sections will be formed and two or more sessions held simultaneously.

It is the desire of the World's Congress Auxiliary that a few addresses of a popular nature shall be given, with a view to stimulating an interest in the development of the science on the part of the public.

T. C. C.

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Extra copies of the articles appearing under the head of Studies for Students will be printed and kept on sale for the use of teachers and advanced classes. The prices will be fixed as low as practicable, and a standing list published in the advertising columns of the JOURNAL.

REVIEWS.

On the Glacial Succession in Europe. By Prof. JAMES GEIKIE. Transactions of the Royal Society of Edinburgh, Vol. XXXVII., Part I. (No. 9), 1892, pp. 127-149 (with a map).

In this timely essay Prof. Geikie reaches the following conclusions:

1. The record of the first glacial epoch is found in the Weyborn Crag of Britain, and the ground moraine beneath the "Lower Diluvium" of the continent. During this epoch, the direction of the ice movement in southern Sweden was from the south-east to the northwest. This first glacial epoch of which direct evidence is adduced was followed by an interglacial interval, during which the forest-bed of Cromer, the breccia of Hötting, the lignites of Leffe and Pianico, and certain beds in central France were deposited. During this interglacial epoch, the climate is believed to have been very mild.

2. There followed a second epoch of glaciation, when the ice sheet of Britain became confluent with that of the continent. This was the epoch during which the ice sheet reached its southernmost extension. Its depositions are found in the lower boulder clays of Britain, the lower diluvium of Scandinavia and north Germany (in part), the lower glacial deposits of south Germany and central Russia, the ground moraines and high level gravel terraces of Alpine lands, and the terminal moraines of the outer zone. During this second glacial epoch, Alpine glaciers are believed to have attained their greatest development. This epoch of extreme glaciation was followed by an interglacial interval, during which Britain is believed to have been joined to the continent. During this interval, the climate became temperate. In Russia (near Moscow) there seems to be evidence that it was milder and more humid than that of the same region at the present day. Toward the close of the mild epoch, submergence seems to have been accompanied by an increasing degree of cold, which finally ended in another glacial epoch.

3. The subsidence which marked the close of the second interglacial interval, marked likewise the inauguration of the third glacial epoch. Its work is represented in Britain by the upper boulder clay, in Scandinavia and Germany by the lower diluvium (in part), in central Russia by the upper glacial series, in Alpine lands by ground moraines and gravel terraces. The ice sheets of Scandinavia and Britain were again confluent, but did not extend quite so far south as during the second glacial epoch. This third glacial epoch is believed to have been followed by another interglacial interval, during which fresh water alluvia, lignite and peat accumulations were made. These are represented by the interglacial beds of north Germany, and by some of the so-called post-glacial alluvia of Britain. There were also marine deposits on the coasts of Britain and on the borders of the Baltic. During this interglacial interval, Britain is believed to have been continental. The climate was temperate, but in the course of time became more severe. This increasing severity seems to have been accompanied by submergence, which amounted to something like 100 ft. below the present sea-level on the coasts of Scotland. The Baltic provinces of Germany were also invaded by the waters of the North Sea.

4. There followed a fourth period of glaciation, during which the major part of the Scottish Highland was covered by an ice sheet. Local ice sheets existed in the southern uplands of Scotland and in mountain districts in other parts of Britain, and the great valley glaciers sometimes coalesced on the low lands. Icebergs floated out at the mouths of some of the highland sea-lochs. In some places, terminal moraines were deposited upon marine beds which were then in process of formation. These beds are now 100 ft. above the sea level. At this time Scandinavia was covered by a great ice sheet, which yielded icebergs to the sea along the whole west coast of Norway. The ground moraines and terminal moraines of the mountain regions of Britain represent the deposits of this ice epoch. The upper diluvium of Scandinavia, Finland, and north Germany represent the work of the contemporaneous, but not confluent, ice sheet of the continent. In the Alps, terminal moraines in the large longitudinal valleys were made at the same time.

This fourth glacial epoch was followed by a fourth interglacial interval, during which fresh water alluvial deposits were made, and also the "lower buried forest and peat" of Britain and northwestern Europe. At this time,

Scotland seems to have stood 45 to 50 feet lower than now, and Carse clays and raised beaches represent the work of the sea. During this interglacial interval, Britain is believed to have become again continental, while the climate became so far ameliorated as to allow the growth of great forests. Subsequently the insulation of Britain was effected, and this was followed by a climate which was probably colder than the present.

5. The severity of the climate which marked the close of the fourth interglacial interval was such as to bring about local glaciation in some of the mountain valleys of Britain. Here and there the glaciers projected their moraines so far down the mountains that they rest on what is now the 45 to 50 feet beach. In the Alps, this fifth epoch of glaciation is represented by the so-called post-glacial moraines in the upper valleys. This is believed to have been the last appearance of glaciers in Britain. The dissolution of these glaciers was again followed by an emergence of the island, and by more genial climatic conditions.

In support of his conclusions, Prof. Geikie cites some striking facts which are not so widely known as they should be. For example, Swedish geologists have found evidences that there was an ice sheet antedating that which deposited the "lower diluvium," and that during this earlier glaciation the direction of ice movement in southern Sweden was from the south-east to the north-west. The ground moraine deposited by this ice sheet is overlain by the "lower diluvium" which was produced by an ice movement from the north north-east to the south south-west, or nearly at right angles to the first. Again, near Moscow, there exist interglacial beds whose plant remains indicate a climate milder and more humid than that of the present time. These interglacial beds, it will be observed, occur in the region of the "lower diluvium" quite beyond the margin of the ice which produced the "upper diluvium" of Germany and Scandinavia. During this interglacial interval, Prof. Geikie maintains that no part of Russia could have been covered with ice. If, then, within the limits of the area covered by the "lower diluvium," and not by the "upper," distinct beds of glacial drift are separated by such beds as those cited, there can be no question but that such separation marks two distinct glacial epochs. If there was an earlier glaciation when the movement of the ice in Sweden was at right angles to that during which the lower part of the "lower diluvium" was produced, this also would seem to be good evidence of three ice epochs prior to the "upper

diluvium." The epoch of the "upper diluvium" would then constitute the fourth glacial epoch, and this is the interpretation of Prof. Geikie.

Outside the area of the European continental ice sheet, facts are adduced in striking confirmation of the multiple ice epoch theory. These facts are found in Switzerland, where evidences of multiple glaciation have been recognized, and in the Pyrenees where evidences of three separate ice epochs have been found. In France, evidences of an interglacial interval have been found in the region of the Puy de Dôme of such duration as to allow the excavation of valleys to a depth of 900 feet. The length of time which would be required for such stupendous erosion must certainly be regarded as sufficient to allow the preceding and succeeding glaciations to be considered as belonging to two distinct epochs.

Another point of great significance and interest which Prof. Geikie's essay brings out, is the correlation in Britain between epochs of glaciation and epochs of subsidence on the one hand, and between interglacial intervals and epochs of elevation on the other. If Prof. Geikie's interpretation be well founded, and so far as we are able to judge from the facts presented this is the case, his conclusions would seem to be fatal to the hypothesis that glacial climate was produced by northern elevation.

The map which Prof. Geikie gives, showing the limit of ice advance during the fourth glacial epoch, seems to us open to criticism. On the ground of personal observation, the writer believes that the ice sheet of the glacial epoch here represented did not extend notably, if at all, beyond the Baltic Ridge. [\[18\]](#)

[\[18\]](#) See *American Journal of Science*, May, 1887. In a recent letter, Prof. Geikie indicates that he is convinced, from subsequent personal observation, that his map is erroneous so far as the limit of the ice of this epoch is concerned. The mapping given was based on the opinion of others.

Prof. Geikie is an advocate of Dr. Croll's astronomical theory of glacial climate, and thinks that even five is not the full number of glacial epochs belonging to the Pleistocene period. He believes there may have been a series of glacial epochs increasing in severity to a maximum represented by what is now designated as the second glacial epoch. This maximum was followed by a series of epochs of diminishing severity, represented by what

he designates the third, fourth and fifth epochs. The essay is a timely contribution to glacial geology.

ROLLIN D. SALISBURY.

ANALYTICAL ABSTRACTS OF CURRENT LITERATURE. [\[19\]](#)

[\[19\]](#) Abstracts in this number are prepared by Henry B. Kummel, Chas. E. Peet, J. A. Bownocker.

The Sub-Glacial Origin of Certain Eskers. By WILLIAM MORRIS DAVIS, Harvard University. (Proceedings of the Boston Society of Natural History, Vol. XXV., May 18, 1892).

A critical discussion of the conditions under which it is conceived certain eskers and sand plateaus (plains) were formed. The Auburndale district, ten miles east of Boston, presents three classes of modified drift deposits;—sand plateaus, eskers, and kames. These deposits are well exposed.

The sand plateaus have the characteristics of delta deposits of glacial streams,—even surfaces, well-bedded sands and gravels, the beds sloping outward from the "head" at an angle of 12° to 20°, and in close agreement with the slope of the plateau front, a lobate margin, deposits distinctly coarser at the head than near the front, and a series of nearly horizontal roughly cross-bedded gravels overlying the sloping beds.

The eskers are essentially of the same material as that of the plateau, often so poorly stratified as to render differentiation of the beds difficult. The interstices between the pebbles are often unfilled, although there is abundance of fine material in adjoining layers. This "open work" is taken to indicate rapid deposition, and seems to preclude the supposition that the gravels have settled down from a superglacial position, or been traversed by currents of any volume. In several instances the eskers can be followed to direct union with sand plateaus. Towards its lower end the esker frequently "gives out branches" and "the adjacent lowland surface becomes more or less encumbered with sand mounds or kames," indicating a decayed margin of the ice.

Prof. Davis' conclusions are:

"1. The eskers and sand plateaus of Auburndale and Newtonville were formed by running water just inside and outside of the ice margin in the closing stage of the last glacial epoch.

"2. The ice-sheet was a stagnant, decaying mass at the time of their formation, as is shown by the ragged outline of its margin.

"3. Eskers and sand plateaus are genetically connected; the term, feeding-esker, is fully warranted by the relation of the two in position, structure, and composition.

"4. The sand plateaus were made rapidly; this is proved by the absence of disordered beds at their heads, where space would have been opened by the backward melting of the ice had the forward growth of the plateau been slow. The eskers were also made rapidly, as is shown by their 'open-work gravels.'

"5. The diversion of the feeding streams to other outlets left the plateaus and the eskers without further energetic action as the ice melted away from them.

"6. The present form and structure of the eskers are more accordant with the supposition of a subglacial origin than of a superglacial origin; but it is not intended to imply that other eskers of more irregular form and different structure could not have been deposited in superglacial channels."

H. B. K.

Studies in Structural Geology. By BAILEY WILLIS, U. S. Geol. Surv. (Transactions of the American Institute of Mining Engineers, June, 1892).

The paper aims "to present some of the results of observation of the geologists of the Appalachian division during the past three years on the subject of structural geology in the Appalachian province." The structural features are all of one type but of different phases, comprised in four great districts. 1) the district of close folding, 2) a district whose chief structural

characteristic is cleavage, 3) a district of open folding, 4) a district of faulting and folding. The answer to the questions, Why did the strata bend in the district of open folding, and why did they break in the district of faulting, is that the thrust affected them according to their rigidity under their respective conditions of superincumbent load. "We know that load up to a certain point restrains fracture in material under thrust." In the district of open folding the Devonian limestone is the most rigid of the strata and "the one which would most effectively transmit the compressing thrust and would control the resulting structure." In the district of open folding this limestone was prevented from breaking and faulting by a load of superincumbent strata exerting a pressure of 10,000 to 23,000 pounds per square inch, while in the faulted district a load of 5,000 to 10,000 pounds per square inch permitted the strata to break and fault.

The answer to the question, Why did the compression affect this zone, is given. "It becomes apparent on study of sections that where compression raised a great arch there previously existed a bend from a nearly horizontal to a descending position in the principal stratum transmitting the thrust. Greater anticlines and synclines originated in upward and downward convexity of initial dips, due to unequal deposits of sediments which depress underlying strata in proportion to their weight. Such folds may be called original." The Pottsville, Mahanoy, Shamokin and Wyoming coal basins of Pennsylvania belong to this class.

Experiments have recently been carried on in the office of the United States Geological Survey reproducing the different forms of folding. The experiments differed from other experiments in that 1) the materials used to simulate the stratified rocks varied in consistency from brittle to plastic, according to the depth at which deformation is supposed to take place; 2) the compression was exerted under a movable load representing the weight of superincumbent strata; 3) the strata rested on a yielding base to simulate the condition of support of any arc of the earth's crust. The following are the conclusions from the experiments:

1. "When a thrust tangentially affects a stratified mass, it is transmitted in the direction of the strata, and by each stratum according to its inflexibility. At any bend the force is resolved into components, one radial,

the other tangential to the dip beyond the bend; the radial component, if directed downward, tends to depress the stratum and displace its support.

2. "A thrust so resolved can only raise an anticline or arch which is strong enough to sustain the load lifted by its development; such an arch may be called competent; and since strength is a function of the proportions of a structure, it follows that, for a given stratum, the size of a competent anticline will vary inversely as the load; or for a given load the size will vary as the thickness of the effective stratum.

3. "The superincumbent load borne by a competent anticline is transferred to the supports of the arch at the points of inflection of the limbs.

4. "When a competent arch is raised by thrust from one side, the load transferred may so depress the resulting syncline further from the force that an initial dip will be produced in otherwise undisturbed strata; this dip will rise to a bend from which a new anticline may be developed. This anticline is a result of the first, and may be called 'subsequent' in distinction to original folds. Since subsequent folds are simply competent structures, their size will be determined by conditions of thickness and load, and for like conditions they should be equal; and they must, in consequence of conditions of development, be parallel to the original fold and to each other. An example of an original fold with its subsequent anticlines is the Nittany arch and the group of parallel anticlines which lie southeast of it, extending northeast from the Broad Top basin."

C. E. P.

The Catskill Delta in the Post-Glacial Hudson Estuary. By WILLIAM MORRIS DAVIS. (From the Proceedings of the Boston Society of Natural History, Vol. XXV., 1891).

The post-Tertiary trenches of the Hudson and its tributaries are in the main filled with clay beds, which, covered by a thin deposit of sand, rise in terraces 130, 150, or even 180 feet above tide-water. These clays are the

result of a late glacial or post-glacial submergence of the valley, but their upper surface does not indicate the amount of their submergence, as they are bottom deposits. Delta deposits made by the tributary streams, where they entered the Hudson estuary, would indicate the amount of submergence.

Such deposits are found on the Catskill a mile north of Cairo, and eroded remnants are traceable for three or four miles down stream. The surface is characterized by great numbers of water-worn stones up to fifteen or eighteen inches in diameter. The lobate margin, where present, is poorly defined. These deposits range from 290 feet (aneroid) above tide, up river, to 270 feet further down. One-tenth of a cubic mile of material seems to have been washed into the Catskill trench at the point of this delta between the time of the ice departure and the elevation of the land. Subsequent terracing has removed half that amount.

The course of the Catskill at Leeds, where it crosses a ledge of hard Corniferous limestone is probably of post-glacial superimposed origin, but the preglacial valley cannot be definitely fixed.

H. B. K.

Geological Survey of Alabama.—Bulletin 4. By C. WILLARD HAYES.
(Report of the Geology of Northeastern Alabama and Adjacent Portions of Georgia and Tennessee).

This report covers an area of 5950 miles, two-thirds in Alabama. Topographically it falls into three divisions: 1) the Cumberland and other plateaus of the northwest; 2) in the center, anticlinal valleys—Browns and Wills, with the synclinal mountains—Sand and Lookout; 3) the monoclinical mountains, the "flatwoods" (Coosa shales) and the chert hills (Knox limestone) of the southeast. The drainage of the first is radial from the center of the plateau to the Tennessee; that of the second, once consequent upon the folded structure, is now adjusted to the strike of the soft beds.

The formations are Cambrian, Silurian, Devonian and Carboniferous. Total thickness is from 13,000 to 18,000 feet in the east, but decreases westward. Hard sandstones of the Carboniferous form the cappings of the plateaus and synclinal mountains. In the anticlinal and monoclinal valleys the Silurian and Cambrian appear. The rocks pass from the nearly horizontal beds of the plateau region, by narrow unsymmetrical anticlines with steeper dip on the northwest side, and by broad shallow synclines, to the complicated folds of the southeast. The axes of these latter folds dip more or less abruptly northward and southward, causing the ridges to assume zigzag courses. Synclines are often crossed by anticlines.

Thrust faults exist, some of great magnitude, and traceable for 200 to 300 miles. By the "Rome thrust fault" the Cambrian shales have been shoved four to five miles over upon the Carboniferous shales. Most of the overthrust strata have been worn away, but tongues of Cambrian shale still remain to all appearances lying conformably upon the Carboniferous strata. Transverse thrust faults terminate Gaylor's ridge, Dirt Seller Mountain, and Lookout Mountain on the south.

H. B. K.

The Correlation of Moraines with Raised Beaches of Lake Erie. By FRANK LEVERETT, U. S. Geol. Surv. (Wisconsin Academy of Science. Vol. VIII., 1891).

References have been made in Geological literature to the beaches of the eastern portion of the Lake Erie basin, but up to the time of Mr. Leverett's work none of the beaches had been completely traced. Mr. Gilbert had discovered that several of the raised beaches do not completely encircle Lake Erie, and supposed that their eastern termini represent the successive positions of the front of the continental glacier during its retreat northeastward across the Lake Erie basin. Mr. Leverett verifies this theory by demonstrating that certain moraines are the correlatives of the beaches. They are as follows:

I. The Van Wert or upper beach and its correlative moraine, the Blanchard ridge. II. The Leipsic or second beach and its correlative moraines. III. The Belmore, or third beach and its correlative moraine.

I. The Van Wert beach extends eastward from the former southwestward outlet of Lake Erie near Fort Wayne, Indiana, to Findlay, Ohio, where it joins the Blanchard moraine. Through Indiana and Ohio its altitude is quite uniformly 210 feet above Lake Erie.

While the Van Wert beach was forming, the ice front was the northeastern shore of the lake as far east as Findlay, Ohio, its position being marked by the Blanchard moraine. East of Findlay, where the Van Wert beach joins it, the moraine is of the normal type. But west of Findlay, it presents peculiarities of topography and structure, resulting from the presence of lake water beneath the ice margin. The water was shallow and incapable of buoying up the ice-sheet, and producing icebergs. The motion of the water under the ice-sheet produced a variable structure. This is the only instance of a moraine demonstrably formed in lake water.

II. The Leipsic, or second beach, was formed after the ice had retreated from its position marked by the Blanchard moraine. Its altitude is 195 to 200 feet above Lake Erie. It has its terminus near Cleveland, where it connects with the western end of a moraine.

III. The Belmore beach and its correlative moraine. Between the Leipsic beach and the present shore of Lake Erie are several beaches. One of these, the Belmore beach, terminates near Cleveland, while the others extend into southwestern New York, and probably connect with moraines, though this connection has not been traced. The general altitude of the Belmore beach in Ohio is 160 to 170 feet above Lake Erie. Unlike the Van Wert and Leipsic beaches, it does not directly connect with a moraine at its eastern end, but a gap of ten miles intervenes. Terraces at Cleveland, Mr. Leverett thinks, make a connection between the eastern end of the beach and the western end of the moraine at Euclid, Ohio.

C. E. P.

The Climate of Europe During the Glacial Epoch. By CLEMENT REID.
(Natural Science. Vol. I, No. 6, 1892).

Temperature of the Sea.—The temperature of the English Channel was similar to that where the isotherm of 32° F. is now situated. The winter temperature can scarcely have been 20° colder than at present. The Mediterranean was perhaps 5° colder than now.

Temperature of the Land (air).—It does not appear that the climate of the lowlands of southern Europe can have been 20° lower than the present mean; 10° or perhaps less appear to have been the refrigeration in the Mediterranean region. The temperature at the southern margin of the ice-sheet was about 20° colder than at present. The temperature increased rapidly towards the south. Recent observations seem to show that throughout central Europe there was a period of *dry* cold, causing the country to resemble the arid regions of central Asia.

J. A. B.

On the Glacial Period and the Earth-Movement Hypothesis. By JAMES GEIKIE, Edinburgh, Scotland. (Read before the Victoria Institute, London).

Geologists generally admit that there have been at least two glacial epochs, separated by one well-marked interglacial period. The closing stage of the Pleistocene period was one of cold conditions in northwestern Europe, accompanied by land depressions. After this came a genial climate with a union of the British islands among themselves and also with the continent. This was followed by a cold, humid condition.

Upham maintains that the whole of North America north of the Gulf of Mexico stood at least three thousand feet higher at the beginning of the glacial epoch than at present. Fiords were formed before glacial times and so can not be cited as evidence of high land during the glacial period. An elevation of land in the northern part of North America and Europe could not produce glaciation in their southern parts. The deflection of the Gulf

Stream by the sinking of the Panama, Professor Geikie argues, could not produce the conditions which prevailed during the glacial epoch. The Earth-Movement hypothesis, he believes, accounts neither for the widespread phenomena of the ice-age, nor for the remarkable interglacial climates. Some maintain that the warm interglacial period was produced by the rise of the Panama land, the sinking of the lands to the north, and the turning of the Gulf Stream from the Pacific into the Atlantic. Why then, asks Professor Geikie, do we not have such a climate now?

J. A. B.



ACKNOWLEDGMENTS.

The following papers have been donated to the library of the Geological Department of the University of Chicago, mainly by their authors:

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